

Funding Global Public Good Institutions via taxes on aviation emissions- a comparison of three proposed mechanisms

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Abstract

The EU is considering taking unilateral action to achieve higher carbon pricing for international aviation. It has been proposed that the EU could start a club mechanism whereby participating countries would have to tax at a fixed rate the emissions of all outgoing flights as well as all flights coming in from non-participants. They would then have to allocate a certain proportion of the tax revenue to some agreed-upon global institution for climate change mitigation. I propose an alternative club mechanism in which participating countries could allocate their tax revenue to Global Public Good Institutions (GPGIs) of their choice from a set of GPGIs recognised as eligible in the treaty establishing the mechanism. Of the tax revenue that participants decide to not allocate, they could retain a fixed proportion. The remaining revenue would be added to the GPGIs, in proportion to the aggregate allocations made to them. I consider an example where the set of eligible GPGIs consists of 2 GPGIs for global health and 4 for climate change mitigation. I estimate the payoffs that countries would derive from money being added to the different GPGIs, taking into account the heterogeneous benefits generated by each GPI, the rents that some GPGIs create and the redistributive effects that some GPGIs induce by changing e.g. the oil price. Using an iterated best response algorithm and assuming that initially only the EU and China participate, I predict that this new mechanism can achieve 22% more welfare gains than the previously proposed one, raising 50% more revenue for GPGIs. I then enrich the mechanism by allowing countries to also give to so-called “Proportional Matching Funds”, which distribute their budget to subsets of GPGIs, in proportion to the direct allocations made to these GPGIs. This enriched mechanism can achieve 67% more welfare gains than the previously proposed one, raising 100% more revenue for GPGIs.

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1 *Introduction*

In 2016, the member states of the International Civil Aviation Organisation (ICAO) adopted the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA). 81 States, representing 76.63% of international aviation activity, intend to voluntarily participate in CORSIA from its outset. Starting in 2021, the participating countries will oblige airlines to offset a part of their emissions by buying certificates for projects that have been certified to have reduced emissions. CORSIA thus has the potential to improve global welfare in two ways: firstly, causing reductions in aviation emissions due to the carbon price, which will equal the price of offsets. Secondly, through the money generated for the offset projects which will be able to sell their emissions reductions certificates to airlines.

The offset eligibility standard under CORSIA is currently under negotiation. 14 offsetting programs have applied to be eligible. Depending on the standard chosen, different types of projects will get financed, such as REDD+ projects for reducing tropical deforestation or Clean Development Mechanism (CDM) projects for financing renewable energy projects in developing countries. Different offset programs will generate rents for different countries. This can explain why countries have struggled to reach agreement. China has proposed that each state should be allowed to determine which offset standard to accept. However, this would create incentives for countries to choose lenient standards, so as to decrease the burden weighing on the airlines.

In fact, different offset standards differ greatly in their stringency. Warnecke et al. (2019) find that under some offset eligibility rules that have been proposed, demand from CORSIA would be met by existing projects that would continue anyway and the price for the offsets per ton of CO_2 would drop below \$1 (which would mean that the carbon price that airlines would face would correspond to only about 0.3% of ticket prices). Based on this concern, France, the Netherlands, Portugal, Belgium, Austria, Finland, and Norway have threatened independently to withdraw from the agreement if the offset eligibility standards are not chosen stringent enough. The EU is considering instituting further carbon pricing policies for international aviation in parallel to CORSIA. In this study, I will compare 3 proposals for club mechanisms for taxing emissions from international aviation and raising money for Global Public Good Institutions (GPGIs) that groups of countries such as the EU could initiate. By a GPGI I shall mean any institution that uses money to cause increases in the provision of a particular global public good. One example of a GPGI is a fund to buy up carbon offsets for a given certification standard.

The first proposal that I will define, named CORSIA+, is based on suggestions that were commonly made in response to a public consultation held by the EU on the question of which carbon pricing policies for international aviation it should implement. In CORSIA+, participating countries would be obliged to tax at an agreed-upon rate x all emissions from outgoing international flights, as well as all emissions from flights coming in from non-participating countries. Each participating country would be allowed to retain for itself a proportion

r of the tax revenue collected. The value of the retention rate parameter r would be fixed in the treaty establishing CORSIA+. The remaining part of the tax revenue would be allocated to an agreed-upon GPGI for climate change mitigation.

I study the likely outcomes of CORSIA+ for different values of the retention rate parameter r in a model that I also use to analyze the other two mechanisms that I propose. In the model, the world is partitioned into the 12 regional blocks of countries of the RICE model (Nordhaus 2010), each of which is assumed to act as a unitary player maximizing the aggregate payoffs of the countries that it comprises. I focus on the case of a small tax rate x that CORSIA+ would oblige participants to impose on emissions from international flights and estimate the countries' payoffs for this case. The assumption of x being small is made to simplify the model, allowing me to make several linearity assumptions. To estimate the payoffs in the game, I use data and econometric estimates where possible, whilst resorting to several ad hoc assumptions where this appears necessary.

To predict the eventual outcome of the game induced by CORSIA+ that would result if initially only China and the EU participate, I use an iterated best response algorithm. At each step of the algorithm, a randomly selected player is given the opportunity to adjust his strategy to play a best response to the other players' strategy profile at that step. For CORSIA+ the algorithm always converges to a Nash equilibrium, which is taken to be the prediction for the eventual outcome of the game induced by the mechanism. For CORSIA+ I find that for $r \leq 0.55$, the algorithm always converges to a Nash equilibrium with no participation and for $r \geq 0.6$ it always converges to a Nash equilibrium with full participation. The question arises as to whether we could modify CORSIA+ so as to enhance participation incentives in other ways than by increasing r . The alternative mechanisms that I propose in this article are motivated by this question and answer it in the affirmative.

The first alternative proposal is called the "Simple MGF Mechanism", where MGF stands for "Multiple Global Public Good Institution Financing". In this mechanism, participants would not be obliged to allocate money to a single fixed GPGI, as is the case in CORSIA and CORSIA+. Instead, there would be a set S of GPGIs that would be recognized as being eligible for receiving money through the mechanism. Participating countries could allocate the tax revenue that they collect to any GPGIs defined as eligible in the treaty. Of the tax revenue that they decide to not directly allocate to GPGIs, they could retain a fixed proportion r , where the retention rate parameter r would be fixed in the treaty. The remaining proportion, $1 - r$, of the money not directly allocated to GPGIs would be added to the GPGIs in proportion to the direct allocations made to them, thus enhancing the influence of those that make direct allocations to GPGIs.

In the numerical simulations, I find that the Simple MGF mechanism can achieve 22% more gains in expected welfare than CORSIA+. At the value of r that maximises the expected welfare for the Simple MGF mechanism, 50% more money is raised for GPGIs than under CORSIA+ with the corresponding

value of r .

To further enhance the incentives to participate and to allocate revenue to GPGIs, I enrich this mechanism by allowing countries to give to so-called Proportional Matching Funds (PMFs). For each subset J of size at least 2 of the set S of eligible GPGIs, there would be a corresponding PMF. A PMF for a set J of GPGIs would distribute the money it has received to the GPGIs in J , in proportion to the direct allocations that these GPGIs receive through the mechanism. In this enriched mechanism, called “MGF Mechanism with PMFs”, the money that is not directly allocated serves to subsidize the direct allocations made to the PMFs.

In the numerical simulations, the MGF mechanism with PMFs achieves higher expected welfare than the other mechanisms. If one sets the retention rate parameter for each mechanism so as to maximise expected welfare, one finds the following: The MGF with PMFs achieves 67% more welfare gains than CORSIA+ and raises 100% more tax revenue than CORSIA+.

The article is organised as follows. In section 2, I review some of the literature on the broader question that this study contributes to: How to design mechanisms to increase the provision of global public goods? Then I provide some background information on CORSIA in section 3. In section 4 I report some suggestions for unilateral policies to price emissions from international aviation that were made in response to a public consultation held by the EU. Based on these suggestions, I define CORSIA+. Then, in section 5, I describe the model with which I analyze the different proposed mechanisms. In section 6 I define the MGF mechanisms. I briefly describe 6 GPGIs that contribute in different ways to climate change mitigation and global health. The numerical simulations are done, assuming that the set of eligible GPGIs for the MGF mechanisms consists of these 6 GPGIs. Estimates for the payoffs that the players derive from these GPGIs is provided in 6.3. The detailed description of the methods used to compute these estimates is given in appendix B. Section 7 starts by precisely defining and motivating the algorithm used to predict the eventual outcomes of the games induced by the mechanisms. I then provide summaries of the simulation results. Section 8 concludes.

2 Related literature

Nordhaus (2015) proposes a club mechanism to increase carbon pricing around the world. Members in the club would be obliged to establish a carbon price of at least some agreed-upon level and to apply punitive ad-valorem tariffs on all goods that are imported from non-participating countries. Nordhaus emphasizes the risk that such a mechanism could pose to the world trading system. Non-participants might retaliate, which could lead to a trade war.

It would, therefore, be valuable if mechanisms could be designed that would increase the incentives for global public good provision without resorting to punitive sanctions on non-participants. One potential source for such incentives that has been studied in the literature (see Barrett (2005) for a survey) is the

following: treaties can restructure the strategic interaction between countries, transforming it into a three-stage game where in the first stage countries decide whether to participate, in the second stage participants act jointly and in the third stage non-participants play a best response. The more countries participate, the greater the contributions decided in the second stage. As a result, participation incentives emerge.

Incorporating this logic into a model of the Paris agreement’s pledge and review bargaining, Harstad (2018) finds that the level of emissions abatement attainable through this is rather limited. One way to enhance incentives that he proposes is to use conditional matching offers, where countries would commit to abating more if the others abate more. If these matching offers are credible, then the incentives for each country to abate are enhanced because by doing so it will be able to cause others to abate more. Harstad (2016) notes that negotiating conditional offers are time-consuming and costly. Furthermore, he finds that a series of short-term commitment periods lead to a hold-up problem with under-investments in green technology. Based on this, he proposes that conditional pledges should be expressed according to pre-specified formulas.

Conditional matching for contributions on a single dimension (e.g. national emissions reduction) with pre-specified formulas has been studied, amongst others, by Guttman’s (1978, 1987), Guttman (1987), Rübbelke (2006) and Boadway et al. (2011) in a two-stage game: Emissions abatement is divided into an unconditional and a conditional abatement. Countries simultaneously declare matching factors, which define how much conditional abatement they will contribute for a unit of unconditional abatement contributed by each of the other countries. Assuming countries can commit to the matching factors they declare, this scheme allows countries to alter the incentives of other countries. Molina et al. (2019) rigorously prove the remarkable result that in the case of two countries there is a unique subgame perfect Nash equilibrium in this game and at this equilibrium efficient levels of abatement are reached.

One practical difficulty for the implementation of such conditional matching mechanisms is that abatement measures might be difficult to compare. Moreover, it might not be credible that countries will follow through on their commitments. This might necessitate using the threat of sanctions to make the commitments credible (Harstad (2016))¹. In contrast to this, in the MGF mechanisms that we will propose, incentives to contribute money to GPGIs are created without any reliance on countries’ ability to commit. On the other hand, the MGF Mechanisms are far less ambitious than the conditional matching mechanisms analyzed by Molina et al. (2019). The amount of cooperation achieved is fundamentally limited by how much money is raised for GPGIs via the aviation emissions taxes. The mechanisms we propose could serve as a potential incremental opportunity to raise money for GPGIs and thereby complement other mechanisms for global public good provision.

¹The use of sanctions for this seems a lot less problematic than applying punitive sanction to non-participants to an agreement. If sanctions are used to help countries to voluntarily commit to following through on matching offers, these sanctions might never actually be applied and yet be effective.

The starting point of the current paper is that global public goods such as climate change mitigation and the prevention of pandemics are all underprovided and will remain so, at least in the near future. From this, it follows that there are two opportunities to incrementally increase global welfare through mechanisms like the ones studied in this article: raising more money for GPGIs and improving the allocative efficiency of the distribution across the different GPGIs of whatever money is raised. The two MGF Mechanisms studied here are ad hoc proposals motivated by these opportunities.

3 Mechanisms to price carbon emissions from international aviation that have been or are being attempted

3.1 The EU’s unilateral inclusion of international aviation in the EU Emissions Trading Scheme (ETS) (measure taken in 2012 and suspended in 2013)

In 2012 the EU extended its emissions trading scheme to include aviation. Airlines have been receiving annual tradable allowances covering a certain level of emissions. Any shortfall between the allocated sum of free emissions allowances and their actual emissions require the purchase of additional allowances. Initially, all flights arriving in or departing from an airport in the EU were included. The ‘equivalent measures policy’ set out in the EU directive 2008/101/EC promised that whenever ‘a third country adopts measures, which have an environmental effect at least equivalent to that of this Directive’, flights from that country arriving in the EU would be exempted from obligations to be covered with emission permits. The directive stated that ‘the Community scheme may serve as a model’ and that member states should ‘encourage third countries to take equivalent measures’.

However, states such as the US, China, India, Russia, and Saudi Arabia opposed the scheme and partly prohibited their airlines from complying with it (Murphy et al 2016). The Chinese ambassador to the EU suggested that Chinese airlines would no longer buy Airbus planes. In response to this opposition, the EU modified after only one year (in November 2012) the system to a ‘reduced scope’, which means that only flights between and within EU member states are included. This ‘stop-the-clock’ approach is temporary and if no new decision is made, ‘full scope’ will start again in 2024 (Larsson et al 2018), meaning that, as in 2012, all flights arriving or departing in an EU country would again be covered.

3.2 The CORSIA treaty

In 2016, the International Civil Aviation Organization (ICAO) adopted a global scheme requiring airline operators to offset increases in carbon dioxide emis-

sions from international flights above the aggregate emissions in 2020. It is called CORSIA, which stands for “Carbon Offsetting and Reduction Scheme for International Aviation”. The offset obligations only apply to flights between participating countries. A major point of contention in the negotiations has been and continues to be the choice of eligibility criteria for the carbon offsets.

The eligibility conditions for offsets will affect which type of projects will get funded. One example of possible offset programs that could be accepted is the Forest Carbon Partnership Facility’s offsets for reduced tropical deforestation. Another is the Clean Development Mechanism (CDM), which has evolved to now finance mostly renewable energy projects in developing countries. Even once an offset category has been accepted, there remain crucial choices about vintage conditions. For example, Warnecke et al. (2019) find that the CDM offset will be effective only if eligibility is limited to new projects or projects that are at risk of discontinuing greenhouse gas abatement without further support. Without such a restriction, they conclude, the offset demand from CORSIA would be met by projects that would continue anyway and the price for the offsets per ton of CO₂ would drop below \$1. Fearing that the standards that end up being chosen will be very lax and, France, the Netherlands, Portugal, Belgium, Austria, Finland, and Norway have threatened independently to withdraw from the agreement if the offset eligibility standards are not chosen stringent enough.

Based on these concerns of a dilution of the level of ambition of CORSIA, the EU has kept open the possibility of taking unilateral measures to increase carbon pricing on international aviation. As mentioned in 3.1, by default, all flights to and from EU countries will again be included in the EU emissions trading scheme in 2024. The EU appears to also be open to alternative mechanisms, as it held a public consultation in 2016 on the question. In the next section, we will build on suggestions made in response to this consultation. Based on the suggestion we will define a proposal for a club mechanism for taxing international aviation emissions and financing a global public good institution for climate change mitigation.

4 The CORSIA+ proposal (based on common suggestions made in the EU’s public consultation in 2016)

In 2016 the EU held a ‘consultation on market-based measures to reduce the climate change impact from international aviation’. In response to this consultation a common suggestion was that the EU should price carbon emissions from all flights, either by reinstating the ETS with full scope and with ‘equivalent measures policies’ as it did in 2012 (see section 3.1), or by introducing analogous carbon taxes². A second important suggestion was that this time the EU countries should not keep all the money from the permit auctions (or the tax

²France, the Netherlands and Belgium have also recently argued for such European carbon taxes

revenue) for themselves but instead allocate at least a proportion of it to the Green Climate Fund.

The CORSIA+ proposal defined below extends these suggestions into a club mechanism. The countries initiating such a club would agree on a Global Public Good Institution for climate change mitigation for which money will be raised through the mechanism. By the term 'Global Public Good Institution' (GPGIs) we shall mean any global institution that uses money to cause increases in the provision of a specified global public good. The Green Climate Fund fits this definition since it uses its money for climate change mitigation. Another example of a GPGI for climate change mitigation is any fund that uses its available money to buy up carbon offset certificates.

Definition. Proposal 1: The CORSIA+ mechanism

The CORSIA+ mechanism for a given agreed-upon GPGI for climate change mitigation defines a club through the following two obligations that participating countries have to comply with.³:

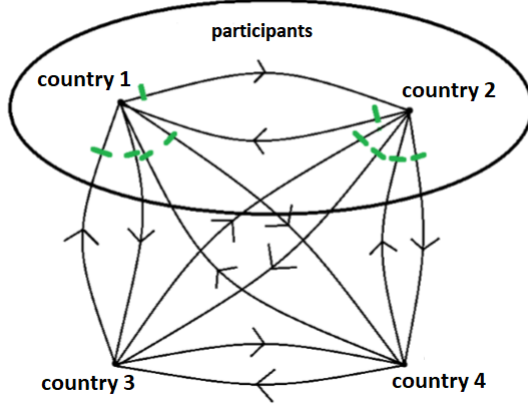
1) **(the taxation obligation)** Each participating country $i \in \textit{Participants}$ is required to levy a tax with rate x on carbon emissions from all outgoing international flights and also on incoming international flights arriving from countries that do not participate.⁴

2) **(the allocation obligation)** Each participating country i must allocate a proportion $1 - r$ of the tax revenue that it collects to the agreed-upon GPGI for climate change mitigation. It can retain the remaining proportion r for itself.

An illustration of the taxation obligation for participants The following diagram illustrates the taxation obligation for participants. Depicted is the case of a hypothetical world consisting of four countries at the strategy profile where country 1 and 2 participate whilst country 3 and 4 do not:

³Throughout this article we will define all mechanisms in terms of taxes. However, one can restate each of them in a way that allows countries to use emissions trading schemes instead of taxes to put a price on emissions and to raise revenue from permit auctions instead of from taxes.

⁴This condition 1 corresponds to the EU's "equivalent measures policy" from 2012 explained in 3.1



Here the green strokes near country 1 indicate the flights that country 1 is obliged to tax and similarly for country 2.

To further illustrate the definition, let us compute the flow of tax revenue M_i that a participating country i ends up collecting. We denote by f_{ik} the carbon emissions from international flights from country i to country k and as before. Let I denote the set of all countries and $Participants$ the set of all participating countries. We have:

$$M_i = x \left(\sum_{k \in Participants} f_{ik} + \sum_{k \in I \setminus Participants} (f_{ik} + f_{ki}) \right) \quad (1)$$

The effects of the retention rate parameter r Suppose all countries participate and we decrease r . Then we cause money that otherwise would accrue to the governments to instead go to the GPGI m for climate change mitigation. Due to the public good nature of climate change mitigation, we should expect global welfare to increase. Thus, setting aside considerations about participation incentives, we should set $r = 0$.

However, to create sufficient participation incentives we might need to set r quite high. This is in fact what we will find in the numerical simulations using the model that will be defined in the next section. In the simulations reported in 7 we will need to set r at least at 0.6 to get any participation.

5 The model

The world is assumed to be partitioned into the 12 blocks of the RICE model (Nordhaus 2010), each of which is assumed to behave as a single player: Africa, China, EU, Eurasia, India, Japan, Latin America, Middle East, Other high income countries, Russia, other Asia. The payoff of each of these players is assumed to be the sum of the payoffs of the countries that it comprises. Moreover, we shall assume that country i 's utility is given by:

$$u_i = t_i - c_i x F_i + Y_i + b_i x G + \theta_s^i Z_s \quad (2)$$

For ease of exposition, we have written here the payoff in the context of the CORSIA+ mechanism and assuming that x is small. The payoff is relative to a situation without the CORSIA+ mechanism.

t_i is the transfer that country i receives, i.e. the money that it can retain from the tax revenue collected if country i participates. F_i denotes the aggregate emissions from international flights from or to country i whose emissions are taxed as a result of the mechanism. Denoting by f_{ij} the emissions from flights from i to j , we have $F_i = \sum_{j \in I} f_{ij} + f_{ji}$ if i participates (where I denotes the set of all countries) and $F_i = \sum_{j \in \text{Participants}} f_{ij} + f_{ji}$ if i does not participate. $x F_i$ is thus the aggregate tax revenue from taxes on flights arriving at or departing from country i , collected due to the CORSIA+ mechanism.

c_i measures how country i evaluates the burden associated with the taxation its flights are subject to. We model it as having two components: $c_i = \sigma_i + \eta_i$. Here σ_i accounts for the welfare effect that country i internalises, which is due to the incidence of the taxes born by its citizens and the potential deadweight loss that they incur. η_i captures redistributive effects between countries. One such redistributive effect is that when a country i increases its aviation taxes some tourists switch to traveling to a different country j instead and thereby rents from these tourists accrue to country j instead of to country i .⁵

Symmetrically, when aviation taxes are charged on flights that are neither to nor from country i , there are redistributive effects in favor of country i . We capture these effects in the term Y_i . We will assume the linear separability of all these effects.⁶ Formally, we will assume that $Y_i = \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate}} \gamma_{ijl} x f_{jl}$. Here γ_{ijk} gives the rents from tourism that are shifted to country i as a result of each dollar of taxes that are imposed on flights from country j to country k . In appendix B.1 we will make assumptions about the parameters σ_i, η_i and γ_{ijl} .⁷

$G := \sum_{i \in \text{Participants}} \sum_{j \in I} f_{ij} + \sum_{i \in \text{Participants}} \sum_{j \in I - \text{Participants}} f_{ji}$ and thus $x G$ is the total tax revenue that is collected due to the MGF mechanism. b_i measures the net benefits that country i perceives due to the reduction in emissions caused by the carbon taxes levied on the flights. This includes two effects: firstly, the reduction in climate change and secondly the reduction in the world market price of oil that is caused by the reduction in the demand for oil from the aviation industry. We will estimate these effects in section B.3.

⁵Keen and Strand (2012) highlight another redistributive effect, namely a terms of trade effect: When one country increases its aviation taxes, the world market prices for tourism services increase.

⁶For small values of the tax rate x the additive separability that we are assuming will approximately be correct.

⁷It could be useful to extend the model to one in which the parameters σ_i, η_i , and γ_{ijk} are derived from consumer preferences and the production structure. In this case we would endogenously obtain the dependence of σ_i, η_i , and γ_{ijk} on x and we could compute the welfare maximising (x, r) for the CORSIA+ mechanism. Then we could similarly compute the maximal welfare that the other mechanisms explained below in section 6.4 can generate. However, this is beyond the scope of this article.

Z_s is the total money raised for the GPGI s . Of course, Z_s depends on the strategies chosen by the players. The θ_s^i are constants that correspond to how country i values a dollar going to the GPGI s .

Equation (2) applies for the case of the CORSIA+ mechanism, where a single GPGI s is selected for being financed via the mechanism. In the alternative mechanisms defined in section 6.4, there is not just one but a set S of GPGIs that will be financed. In this case country i 's payoff becomes:

$$u_i = t_i - c_i x F_i + Y_i + b_i x G + \sum_{s \in S} \theta_s^i Z_s \quad (3)$$

For the numerical simulations, we will assume that the set S consists of the 6 GPGIs listed in 6.2. In section B.4 we will compute values for the various parameters, using econometric estimates and data when possible, whilst resorting to ad hoc numerical assumptions for several of the parameters. Then we will simulate the unfolding of the game using an iterated best response algorithm in section 7. Before doing this, we will in the next section motivate and define the three alternative proposals.

6 Two new proposals for mechanisms to raise money for multiple global public good institutions from taxes on aviation emissions

6.1 Motivation

As already pointed out, in the choice of the retention rate parameter r for the CORSIA+ mechanism there is a fundamental trade-off between achieving strong participation incentives and efficiency: Increasing r is good for increasing participation incentives but it comes at a welfare cost, due to the decrease in the amount of money raised for GPGIs. The question therefore arises as to whether we could design alternative mechanisms that could raise for GPGIs a greater proportion of the tax revenue collected. The three proposals that we will define are partly motivated by this question and answer it in the affirmative.

The three proposals build on CORSIA+ by keeping the taxation obligation unchanged. However, in these alternative mechanisms, the allocation obligation will be different. Participating countries can decide to which GPGIs to allocate the tax revenue that they collect. Reflecting this feature, we call the proposals **MGF Mechanisms**, for 'Multiple Global public good institutions Financing Mechanisms'.

In contrast to CORSIA and CORSIA+, the treaty that would establish the MGF mechanisms would not select just a single GPGI to be financed through the mechanism. Instead, it would fix a set S of GPGIs. The participating countries allocate the tax revenue that they collect to the GPGIs from S , according to rules that we will precisely define in 6.4. For the numerical simulations, we will assume that the set S is given by the 6 GPGIs listed in the next subsection.

6.2 Global Public Good Institutions (GPGIs): 6 examples

We shall mean by a 'Global Public Good Institution' (GPGI) any institution that uses money to cause increases in the provision of a particular global public good that it is dedicated to.

Example 1. Fund to buy up Emission Reduction Certificates from the Clean Development Mechanism (CDM) (existing since 2008)

The CDM is the largest carbon offsetting programme available at present. It was established under the Kyoto Protocol.

Example 2. CEPI (The Coalition for Epidemic Preparedness Innovation) (existing since 2017)

CEPI has so far mostly been focussing on advancing vaccines against known threats. Recently, it has started to also fund new platform technologies with the potential to accelerate the development and manufacture of vaccines against previously unknown pathogens.

Example 3. The Global Fund to Fight AIDS, TB, and Malaria (existing since 2002)

This GPGI allocates funding for programs against AIDS, TB, and Malaria. Allocations made to countries are proportional to the disease burden for each of the three diseases and to an index of economic capacity calculated based on per capita GDP. The Global Fund works closely together with another GPGI, called UNITAID, which has been financed through ticket taxes on international flights since 2006.⁸

Example 4. The Forest Carbon Partnership's Carbon Fund (existing since 2011)

This Fund rewards countries with tropical forests as a function of the policies they pursue to conserve their tropical forests. In the future, this institution will also condition its payments on results (amount of tropical forests conserved), as Norway already does in bilateral contracts.

Example 5. ITER (The International Thermonuclear Reactor) (existing since 2007)

ITER is an international nuclear fusion research and engineering megaproject. Its stated goal is to "demonstrate the scientific and technological feasibility of fusion energy for peaceful use".

Example 6. Carbon Pricing Reward Fund (proposed)

This Fund would reward countries for having low per capita emissions as well as for removing their fossil fuel subsidies and taxing carbon throughout their economies. In this article, we will mean by the Carbon Pricing Reward Fund the version of this proposal that is precisely defined in B.4.4.

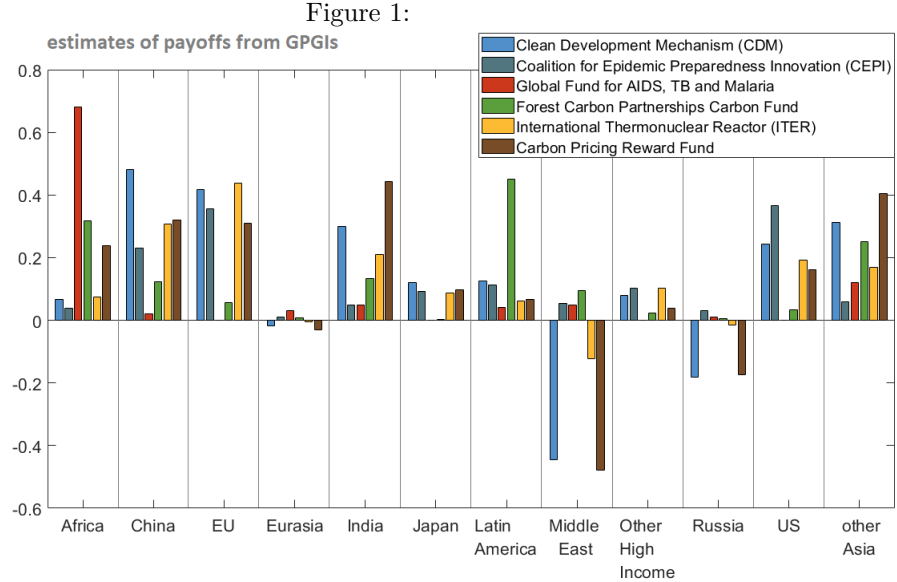
⁸France, Cameroon, Chile, the Republic of Congo, Madagascar, Mali, Mauritius, Niger and Korea have been participating in the financing scheme.

Example 7. Climate change mitigation GPGI assumed to be selected under CORSIA+

As explained in 4, negotiations between the countries participating in CORSIA+ would determine which GPGI for climate change mitigation will be selected for it. Which GPGI would be selected? To form a view about what the likely outcome would be, we can look at the ongoing negotiations under CORSIA, where countries face an analogous question: which offset standards to accept? Under CORSIA, it seems likely that countries will settle on a compromise, where several offsetting standards will be accepted, including both offsets for reducing deforestation as well as offset covered under the Clean Development Mechanism for projects such as renewable energy installation. Based on this, we will assume that under CORSIA+, countries will choose the following GPGI: for each dollar received, it will forward half of it to the fund buying up Emissions Reductions Certificates from the Clean Development Mechanism and half of it to the Forest Carbon Partnership’s Carbon Fund.

6.3 Estimates for the θ_i^s

The model assumes that each country’s payoff depends linearly on the amounts of funding reaching the GPGIs. θ_i^s denotes the utility (expressed in dollars) that player k gains per dollar reaching the GPGI s . Player k ’s utility is assumed to be the sum of the utilities of the countries that it comprises. In B.4 I explain how I estimated the θ_i^s . Figure 1 shows for each player k and each GPGI s the utility, $\sum_{i \in k} \theta_i^s$, that player k derives from a dollar going to s :



I arrive at these numbers using econometric estimates, data, and several ad

hoc assumptions. The methods are fully described in section B.4. However, without going into much detail, we can already provide some explanations of these payoff estimates: Africa most benefits from the Global Fund for AIDS, TB and Malaria, since two-thirds of the Global Fund's disbursements are made to African countries. China benefits substantially from all the GPGIs that reduce fossil fuel use: {CDM, ITER, Carbon Pricing Reward Fund}. The reasons for this are two-fold: Firstly through the reduced climate change and secondly through the decrease in the world market oil price that is caused by the decrease in fossil fuel demand. The latter effect is significant for China since it imports most of its oil. China gains from particularly from the CDM, since a large share of CDM projects are in China, generating rents to the project owners, which we assume the government internalises. Apart from the CDM, the EU has a similar profile of payoffs to that of China.

The gains for the oil importers from reduced world market oil prices come at the expense of oil exporters. This is why the Middle East negatively values money going to the three GPGIs that reduce fossil fuel use. Eurasia's payoff profile is qualitatively similar but much smaller in absolute value since Eurasia is a much smaller player overall.

India and 'other Asia' both greatly benefit from any dollar going to the Carbon Pricing Reward Fund. This is in part because of the reduced climate damages and the reduced world market oil price that result from additional dollars going to the Carbon Pricing Reward Fund. In addition to that, the two players receive reward payments for low per capita emissions and low fossil fuel subsidies from this GPGI and will thereby reap rents.

Latin America benefits greatly from money going to the Forest Carbon Partnership's Carbon Fund. This GPGI rewards countries for conserving tropical forests. Since conservation costs are countries' private information, they reap information rents. Latin America, possessing a large fraction of the world's tropical forests, will reap a large part of these information rents. Other Asia and Africa also value a dollar going the Forest Carbon Partnership's Carbon Fund, again mainly because of the information rents that they get from it.

To summarise: There appear to be three major effects determining the payoffs countries derive from money reaching the GPGIs:

- 1) effects due to the provision of the global public good
- 2) redistributional effects due to induced price changes
- 3) rents

All of these three effects turn out to be heterogeneous across countries. As a result, our payoff estimates, which take into account these three effects, exhibit substantial heterogeneity across countries. The two mechanisms that we will now define are motivated by the hope that it might be possible to harness this heterogeneity to generate participation incentives by allowing participants to affect the allocation of money across GPGIs (in ways determined by the rules of the respective mechanisms).

6.4 Definitions of the two MGF mechanisms

We will now define two new proposals for club mechanisms. We shall call the proposals **MGF Mechanisms** for 'Multiple Global public good institutions Financing Mechanisms'.

6.4.1 Definition of Proposal 2: The Simple MGF mechanism

The Simple MGF mechanism defines a club which any country can join or quit at any point in time. It could be started by any initial subset of countries that would select a set S of *GPGIs*⁹ that would be considered as *eligible* for being financed through the mechanism in a way made precise in the definitions that follow below.

Definition 8. The Simple MGF mechanism for the fixed tax rate x , the fixed set S of GPGIs and the fixed retention rate parameter $r \in [0, 1]$ defines a club through the following two obligations that participating countries have to comply with:

1) **(the taxation obligation)** Each participating country is required to levy a tax with rate x on carbon emissions from all outgoing international flights and also on incoming international flights arriving from countries that do not participate.¹⁰

2) **(the allocation obligation)** As a result of implementing the taxation obligation, each participating country i collects a certain amount M_i of tax revenue. The country can decide on the proportion p_i of M_i that it wants to allocate to GPGIs. It can allocate the amount $p_i M_i$ in any way to the GPGIs from the set S of eligible GPGIs. Of the amount $(1 - p_i) M_i$ that country i does not allocate to GPGIs, it can retain the proportion r (which is the retention rate parameter of the mechanism). The remaining amount, $(1 - r)(1 - p_i) M_i$ is added to the GPGIs proportionally to the aggregate contributions made to them via the mechanism.

For ease of exposition, I have stated the definition here in a static way. In

⁹Throughout this article, we take S to be exogenous. A fully worked-out proposal would need to specify rules for how the set S could be modified over time. One procedure could be the following: At any time participating countries could propose that a GPGI s be added to the set S or removed from the set S . Then all participating countries would cast a vote. These votes could be weighted by population size. The GPGI s would be added to the set S iff the set of participants voting for the addition of the GPGI s comprises at least 50% of the weight of the set of all participants. Alternatively, the weight given to a country could be the sum of its population and some (country-independent) coefficient γ times the aggregate voluntary contributions made by this country to all GPGIs (including not only the GPGIs in S but any GPGI recognized as such by some independent commission) in the 5 years preceding the vote. This would enhance countries' incentives to make voluntary contributions to GPGIs. Thereby, countries could create new GPGIs through voluntary contributions, knowing that if they perform sufficiently well to attract wide support, they will be added to the set S and could receive future funding via the MGF mechanism.

¹⁰This taxation obligation is similar to the EU's "equivalent measures" policy implemented during 2012 based on the EU directive 2008/101/EC.

practice, the treaty establishing the mechanism could allow countries to continually adjust their actions over time.

To clarify the definition, we now restate it with some notation. As introduced in the definition, we shall denote by M_k the flow of tax revenue that a participating country collects by taxing in accordance with the taxation obligation. Each participant has to choose the proportion $p_i \in [0, 1]$ that he wants to allocate to GPGIs. He also has to choose how to distribute the flow of money $p_i M_i$ to the GPGIs. In other words, he has to choose numbers $q_s^i \in [0, 1]$ such that $\sum_{s \in S} q_s^i = 1$ and thereby he allocates the flow $q_s^i p_i M_i$ to the GPI s . These flows of money going to the GPGIs we will call the “direct allocations to the GPGIs”. Of the money $(1 - p_i)M_i$ that country i does not allocate, he retains the amount $r(1 - p_i)M_i$ for himself.

The remaining amount of money, $\sum_{i \in \text{Participants}} (1 - r)(1 - p_i)M_i$, is neither given as direct allocations to GPGIs nor retained by participants. The definition of the mechanism stipulates that this money is to be allocated to the GPGIs in S in proportion to the aggregate direct allocations made to these GPGIs via the mechanism. In other words, from the money $\sum_{i \in \text{Participants}} (1 - r)(1 - p_i)M_i$ the following amount is allocated to the GPI s :

$$\frac{\sum_{i \in \text{Participants}} q_s^i p_i M_i}{\sum_{s \in S} \sum_{i \in \text{Participant}} q_s^i p_i M_i} \sum_{i \in \text{Participants}} (1 - r)(1 - p_i)M_i$$

This concludes the definition of the mechanism.¹¹

Formally, we can write the strategy space for country i as follows:

$$A_i = \{\text{'not participate'}\} \cup \{p_i : p_i \in [0, 1]\} \times \{q^i \in [0, 1]^S : \sum_{s \in S} q_s^i = 1\}$$

6.4.2 A graphical representation of the definition of the Simple MGF mechanism

The MGF mechanism is defined for countries, each of which can decide whether or not to participate and if so, how to participate. However, for the simulations in section 7 we will follow the common practice (see e.g. Nordhaus (2015)) of assuming that countries coalesce in blocks, each of which acts as a unitary player, as we have already stated in the description of the model in section 5. The 12 players are the 12 regions of the RICE model (Nordhaus 2010). The diagram below represents how a strategy profile determines the final allocation of revenue to the GPGIs. The tax revenues that the players collect as a result of complying with the taxation obligation of the mechanism are represented by the white rectangles. In general, each of these tax revenues only depends on what the set of participating players is.

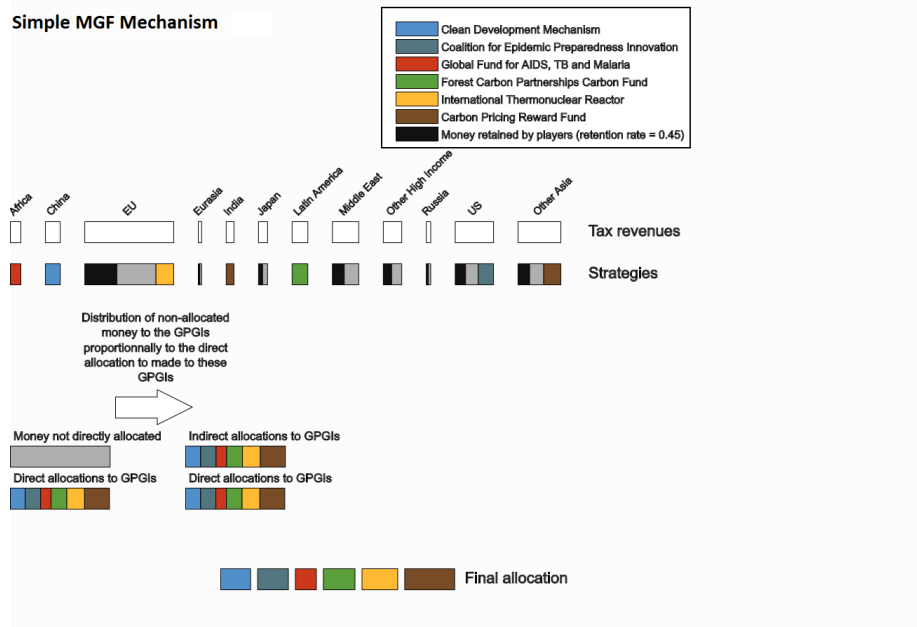
In line with the allocation obligation, each player has to choose what proportion of the tax revenue he collects to allocate to GPGIs. The GPGIs are represented in the diagram by colors as listed in the legend on the top right. For example, we see that Africa allocates all its money to the Global Fund for

¹¹Actually, there is one case that we did not cover in the definition so far: What if all participants choose $q_i = 0$, i.e. if no participant makes any direct allocation to the GPGIs? The details of how we define the rules for this case turn out to be unimportant. Here is the definition that was used for the numerical simulations in 7: If all countries choose $q_i = 0$ then each country i can allocate the amount $(1 - r)M_i$ to the GPGIs.

AIDS, TB and Malaria and China allocates all the tax revenue it collects to the Fund that buys up offset certificates of the Clean Development Mechanism. The EU allocates only a part of the tax revenue that it collects to GPGIs, namely to the International Thermonuclear reactor (ITER), as represented by the partial yellow filling of its rectangle. In line with the allocation obligation, each player can retain a proportion r of the tax revenue that he does not allocate. In the example depicted in the diagram, the retention rate parameter r is 0.45, as indicated in the legend. The revenues that players retain is represented by the black parts of their rectangles. The tax revenues that players neither allocate nor retain are shown in grey.

The lower part of the diagram shows how the final allocations arise from the players' strategies. First, all the money not directly allocated (in grey) is gathered together on the left. Below it are the aggregate direct allocations for each of the GPGIs. In the computational step indicated by the arrow, the money that is not directly allocated is distributed to the GPGIs in proportion to the (aggregate) direct allocations to the GPGIs made through the mechanism¹². By adding together the direct allocations to the GPGIs and the indirect allocations to the GPGIs, we obtain the final allocation of money to GPGIs, which is depicted at the bottom of the diagram.

Figure 2:



¹²Here only the direct allocations made through the mechanism are taken into account. Other contributions, such as voluntary contributions by states or foundations, are not taken into account.

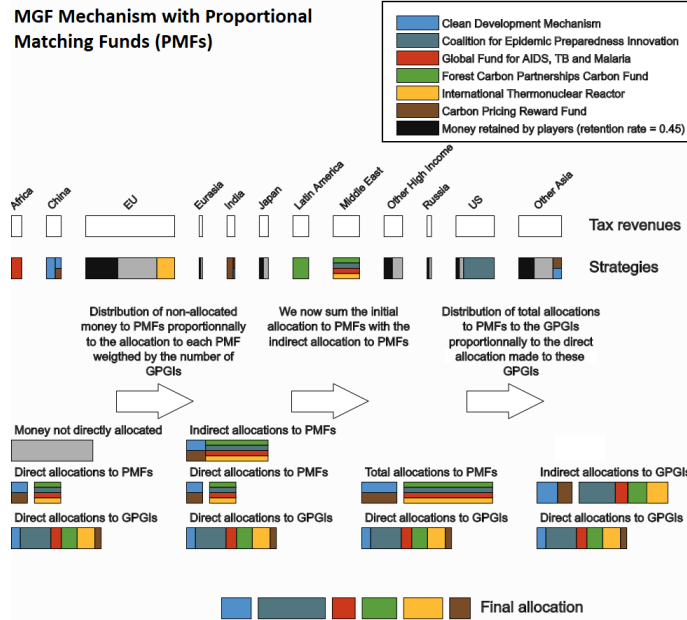
The strategy profile depicted in the diagram is actually a Nash equilibrium that was computed by the iterated best response algorithm described in 7. The EU, US, and Other Asia allocate a part of their tax revenue to their preferred GPGI. Africa, China, and Latin America even allocate all their tax revenue to their respective preferred GPGI. By doing so, each of them causes a part of the money that is not directly allocated to go to their preferred GPGI. For a given value of the retention rate parameter r , this leverage effect enhances these players' participation incentives relative to CORSIA+ where participating players cannot influence the allocation of the revenue to GPGIs (since in CORSIA+ a single GPGI is fixed to which all the revenue that is not retained is allocated). This points to two general effects that might enable the Simple MGF Mechanism to generate more global welfare than CORSIA+: Firstly, for a given value of the retention rate parameter, more revenue might be raised for GPGIs than under CORSIA+. Secondly, for a given value of the retention parameter, there might be stronger participation incentives, which might enable an initial coalition to set the retention rate at a lower value and still induce full participation. A lower retention rate can also increase the amount of revenue raised for GPGIs. Assuming that the algorithms defined in 7 predict the eventual outcome of the game induced by the mechanisms and the payoff estimates reported in 6.3, we will find that both of these effects are indeed substantial.

The question arises as to whether there exist modifications of the Simple MGF mechanism that would further enhance the incentives to participate and to allocate money to GPGIs, without complexifying the mechanism too much. The proposal that I will now define, attempts to achieve exactly that.

6.4.3 Definition of Proposal 3: The MGF mechanism with Proportional Matching Funds (PMFs)

The 'MGF mechanism with Proportional Matching Funds' is an enrichment of the Simple MGF Mechanism obtained by adding Proportional Matching Funds (PMFs) to the set of institutions to which participating countries can allocate the tax revenue that they collect. A PMF is defined by a set J of GPGIs that it "supports". Each PMF is a simple intermediary¹³ that distributes its available money to the GPGIs that it supports, in proportion to the direct allocations made to them through the MGF mechanism, in a way that we will now make precise with the help of the diagram below. The diagram shows how the final allocations are determined in the mechanism, using as an example the case of a strategy profile that actually is an approximate Nash equilibrium (assuming the payoff estimates from 6.3):

¹³An analogous institution already exists in the realm of private philanthropy, where it is sometimes also called 'Proportional Matching Fund' and sometimes called Incentive Fund.



Like in the diagram used to illustrate the definition of the Simple MGF mechanism, the white rectangles represent the tax revenues that the players collect by complying with the taxation obligation (which is the same as in the Simple MGF Mechanism). The direct allocations that players make to the PMFs are represented by parts of the rectangles that are colored with horizontal stripes. For example, China gives a part of its tax revenue directly to the Fund to buy up offset certificates from the Clean Development Mechanism (in light blue) and the other part of its tax revenue to a PMF that supports that same GPGI as well as the Carbon Pricing Reward Fund, which is represented by two horizontal stripes (in light blue and in brown). The Middle East gives all its revenue to the PMF that supports the Coalition for Epidemic Preparedness Innovation, the Global Fund for AIDS, TB and Malaria, the Forest Carbon Partnership's Carbon Fund and the International Thermonuclear Energy Reactor.

The lower part of the diagram represents how the strategy profile determines the final allocation, shown at the bottom of the diagram. On the left, all the money not directly allocated (in grey) is gathered together. Below it, the direct allocations to the PMFs are gathered together. In the particular example, only two PMFs receive positive allocations. Below that, the direct allocations to each GPGI are gathered together.

In the first step (represented by the first arrow), the money that is not directly allocated (in grey) is distributed to the PMFs proportionally to the direct allocations to each PMF, weighted by the number of GPGIs that the PMFs support. We refer to the resulting amounts as the indirect allocations to PMFs. In the next step, we add them to the direct allocations to PMFs, resulting in the total allocations to PMFs.

In the third step, for each PMF, the total allocation that it has received

is distributed to the GPGIs that it supports, proportionally to the direct allocations made to these GPGIs.¹⁴ At the bottom of the diagram, the final allocations to the GPGIs are displayed, which are the sum of the indirect and the direct allocation for each GPI.

Formally, we can write the strategy space for player i as follows:

$$A_i = \{\text{'not participate'}\} \cup \{p_i : p_i \in [0, 1]\} \times \{q^i \in [0, 1]^{\mathbb{P}(S) - \{\emptyset\}} : \sum_{J \in \mathbb{P}(S) - \{\emptyset\}} q_J^i = 1\}$$

As above in the definition of the Simple MGF Mechanism, we denote by p_i the proportion of player i 's tax revenue that he allocates (to GPGIs or to PMFs). $\mathbb{P}(S) - \{\emptyset\}$ is the set of all non-empty subsets of the GPGIs. The interpretation is that for $j \in S$ we have that $p_i q_i^{\{j\}}$ is the proportion of player i 's tax revenue that he allocates directly to the GPI j and for a subset $J \subset S$ with at least two elements, $p_i q_i^J$ is the proportion of player i 's tax revenue that he allocates directly to the PMF supporting the set J of GPGIs.

6.4.4 Motivation

The money in the PMFs enhance the effects that direct allocations to the GPGIs will have. To see this, consider the example of the Global Fund for AIDS, TB, and Malaria in the diagram: Africa is the only player making a direct allocation to this GPI. If Africa did not make this direct allocation, then no money would reach it. By allocating all its revenue to it, Africa also causes a part of the money in the PMF that supports the Coalition for Epidemic Preparedness Innovation, the Global Fund for AIDS, TB and Malaria, the Forest Carbon Partnership's Carbon Fund and the International Thermonuclear Energy Reactor to go to the Global Fund for AIDS, TB and Malaria. Thus money in the PMFs serves to increase the incentives for players like Africa to allocate more money to GPGIs instead of retaining it. This can also increase these players' incentives to participate in the first place.

These beneficial effects of the PMFs provide a rationale for "subsidizing" them. The step represented by the first arrow in the diagram constitutes a way of subsidizing direct allocations to the PMFs: The money not directly allocated is distributed to the PMFs, proportionally to the direct allocations made to the PMFs, weighted by the number of GPGIs supported by each of them. The weighting thus encourages more strongly direct allocations to PMFs that support many GPGIs. The rationale for this is the observation that the above-mentioned beneficial effects of PMFs on the incentives to allocate revenue to GPGIs appear to typically be larger for PMFs supporting many GPGIs.¹⁵

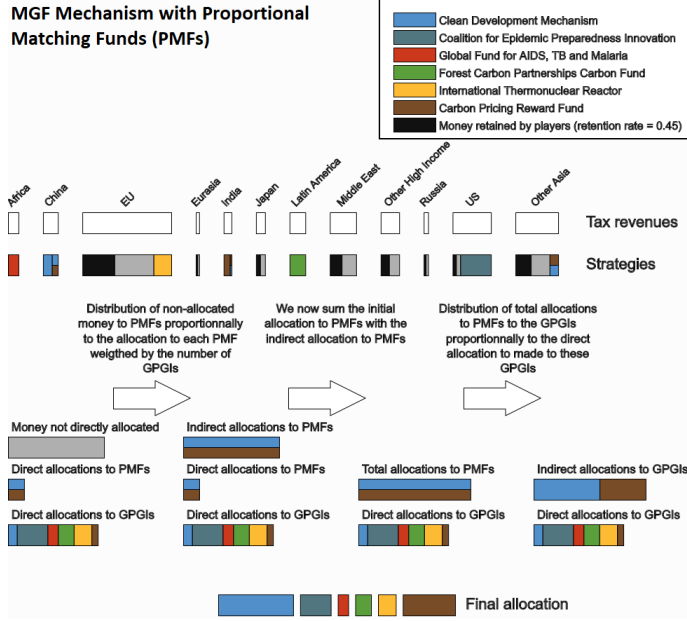
¹⁴Here only the direct allocations made through the mechanism are taken into account. Other contributions, such as voluntary contributions by states or foundations, are not taken into account.

¹⁵The following illustrative example lends support to this claim: Suppose there are 3 GPGIs, denoted X, Y and Z . Let us denote the direct allocations to them by x, y and z . Compare the following two cases:

case 1: there is a PMF supporting X, Y and Z having P_{XYZ} as its available budget. case 2: There are three PMFs, each supporting two GPGIs. Their available budgets are P_{XY}, P_{YZ}, P_{ZX} with $P_{XY} + P_{YZ} + P_{ZX} = P_{XYZ}$.

The subsidisation of PMFs allows those giving to PMFs to greatly affect the final allocation. For example, the diagram below shows what would happen if the Middle East deviates (from the approximate Nash equilibrium shown in the above diagram) by choosing to not allocate any revenue to GPIs or PMFs:

Figure 3:



As a result, now all the money not directly allocated would be added to the PMF supporting the Clean Development Mechanism and the Carbon Pricing Reward Fund. Both of these GPIs would thus get larger final allocations and the other GPIs would receive smaller final allocations. According to the payoff estimates reported in 6.3, the Middle East loses more than \$0.4 for every \$1 going to the Clean Development Mechanism and the Carbon Pricing Reward Fund whilst it benefits from the two GPIs for global health and the Forest Carbon Partnership's Carbon Fund. Overall, the Middle East is best off allocating all

To compare the two cases, let F_X , F_Y and F_Z denote the final allocations. We have:

$$\text{in case 1: } F_X = x + P_{XYZ} \frac{x}{x+y+z}$$

$$\frac{dF_X}{dx} = 1 + P_{XYZ} \frac{y+z}{(x+y+z)^2}$$

$$\text{in case 2: } F_X = x + P_{XY} \frac{x}{x+y} + P_{XZ} \frac{x}{x+z}$$

$$\frac{dF_X}{dx} = 1 + P_{XY} \frac{y}{(x+y)^2} + P_{XZ} \frac{z}{(x+z)^2}$$

Consider the example with complete symmetry, where $x = y = z$ and $P_{XY} = P_{YZ} = P_{ZX} = \frac{1}{3}P_{XYZ}$. Here we have:

$$\text{in case 1: } \frac{dF_X}{dx} = 1 + P_{XYZ} \frac{4}{18}$$

$$\text{in case 2: } \frac{dF_X}{dx} = 1 + P_{XYZ} \frac{3}{18}$$

Thus we see that in this example the marginal incentives to increase direct allocations to GPIs are larger in case 1 than in case 2.

its revenue to the PMF supporting its 4 most preferred GPGIs. The possibility of shifting money away from the GPGIs that affects it negatively enhances the Middle East’s incentive to participate in the mechanism.

6.4.5 Allocative efficiency

I have motivated the MGF Mechanism with Proportional Matching Funds on the grounds that it could enhance participation incentives and increase the amount of money raised for the GPGIs. So far I have left allocative efficiency out of the discussion. In fact, a further motivation for considering the MGF Mechanism with Proportional Matching is that it might lead to a more efficient allocation of money of the GPGIs than what would result under the Simple MGF mechanism, as the following scenario illustrates: Suppose there is a GPGI g that all players value highly but that no player ranks highest. Then in the Simple MGF Mechanism, this GPGI might not receive any money, even if the global marginal payoff from an additional dollar going to this GPGI is higher than that for all of the other GPGIs.¹⁶ In the MGF mechanism with Proportional Matching Funds, some players might have an incentive to give to a PMF supporting g and some other h , given the way that direct allocations to PMFs are subsidised in the mechanism. Some other player preferring g strongly over h might as a result have an incentive to make direct allocations to g , so as to achieve that more of the money in the PMF supporting g and h goes to g .

7 Numerical simulations

For each of the proposed mechanisms, we want to predict the eventual outcome that would result if this mechanism was initiated by a given initial set of players. I use algorithm 3 defined below. However, before proceeding to define algorithm 3, I first discuss two simpler algorithms to motivate the choice of algorithm 3. Each of the algorithms starts with an initial strategy profile and computes a sequence of strategy profiles, using different versions of “iterated best response dynamics”.

Since under CORSIA+ each strategy space is binary, {“participate”, “not participate”}, it is unambiguous what the initial strategy profile is, given a set of initial participants. For the MGF mechanisms, a further assumption is required. I assume that at the initial strategy profile each initial participant allocates all its revenue to its preferred GPGI.

Definition 9. Algorithm 1:

Start with the initial strategy profile.

At each step of the algorithm, select a random player and let him select a strategy that is a best response to the other players’ strategies at that time.

In appendix C I prove in corollary 14, assuming a certain condition on the payoffs that is satisfied for my estimates, that algorithm 1 almost surely con-

¹⁶Brandl et al. (2019) propose alternative mechanisms to mitigate this inefficiency.

verges to a Nash equilibrium. In the simulations, assuming that the initial set of participants is $\{\text{China, EU}\}$, I found that the algorithm quickly converges to always either a Nash equilibrium with no participation or a Nash equilibrium with full participation. For $r = 0.6$, reaching full participation becomes possible, but only at $r = 0.8$ it becomes guaranteed. For $r \in [0.6, 0.75]$, whether full participation is reached depends on whether the EU gets offered the opportunity to change its strategy only after sufficiently many other players have joined.

One can wonder: Suppose the EU and China initiate the mechanism. Would they quickly consider withdrawing from it again, or would they wait and only consider quitting again once it looks like no further players will join? The second theory seems more plausible, which motivates the following alternative algorithm¹⁷:

Definition 10. Algorithm 2:

Start with the initial strategy profile.

phase 1: At each step, select a random player. If this player is not an initial participant, then let him select a strategy that is a best response to the other players' strategies at that time. If this player is an initial participant, then let him select a strategy that is a best response to the other players' strategies at that time, amongst all strategies apart from that of not participating.

If no player has an incentive to deviate, apart from potentially the initial participants wanting to quit, transition to phase 2. Otherwise, continue in phase 1.

phase 2: At each step of the algorithm, select a random player and let him select a strategy that is a best response to the other players' strategies at that time.

In corollary 15 I show, assuming a condition on the payoffs that is satisfied for my payoff estimates, that algorithm 2 generates a unique prediction for the eventual outcome under CORSIA+. For the Simple MGF Mechanism, this uniqueness does not hold, as my simulations will show. Even the convergence of the algorithm is not guaranteed in general. However, for my payoff estimates it does seem to hold, as in all of the runs of algorithm 2 for the Simple MGF Mechanism it converges to a Nash equilibrium.

However, for the MGF Mechanism with PMFs, algorithm 2 appears to typically not converge to a Nash equilibrium in my simulations.¹⁸ One can wonder: Suppose the EU and China initiate the mechanism. Would players continue

¹⁷A further reason for this choice is that the main conclusions of this study, namely that the Simple MGF Mechanism can generate greater expected welfare than CORSIA+ and that the MGF Mechanism with PMFs can generate even larger expected welfare than the Simple MGF Mechanism, appears to hold even more strongly if one assumes that the initial participants always consider quitting.

¹⁸It is well known that iterated best response dynamics can fail to converge, even if there are pure strategy Nash equilibria (Roughgarden 2013). This is indeed the case here: For $r \geq 0.7$ one can show that if all players participate, then no player has an incentive to quit. Since the set of strategies that entail participation is convex, there must therefore exist at least one Nash equilibrium with full participation for all $r > 0.7$. However, in the numerical simulations algorithm 2 appears to never converge for $r \in \{0.7, 0.75, 0.8, 0.85, 0.9, 0.95\}$.

to adjust their strategies, as assumed by algorithm 2? It seems plausible that players would incur some adjustment costs when changing their strategy and would at some point stop adjusting their strategy unless the gains in payoffs from such a change are sufficiently large. This motivates my choice of using algorithm 3, defined below. Algorithm 3 is a modification of algorithm 2 that adds a small adjustment cost that is increased at each step. Players only adjust their strategies if the gains from doing so exceed the adjustment cost.

Definition 11. Algorithm 3:

Start with the initial strategy profile.

phase 1: At each step n , select a random player. If this player is not an initial participant, then let him select a strategy that is a best response to the other players' strategies at that time, taking into account the adjustment cost $a(n)$, where a is a small and slowly increasing function. If this player is an initial participant, then let him select a strategy that is a best response, taking into account the adjustment cost $a(n)$, to the other players' strategies at that time, amongst all strategies apart from that of not participating.

If no player has an incentive to deviate, apart from potentially the initial participants wanting to quit, transition to phase 2. Otherwise, continue in phase 1.

phase 2: At each step of the algorithm, select a random player and let him select a strategy that is a best response to the other players' strategies at that time, taking into account the adjustment cost $a(n)$.

By construction, algorithm 3 always converges to an approximate Nash equilibrium. I take the strategy profile to which algorithm 3 converges, as a probabilistic prediction of the "eventual outcome" of the game.

For the adjustment cost function $a(n)$ I chose a quadratic function. Its coefficients were set sufficiently low so that for CORSIA+ and the Simple MGF Mechanism, algorithm 3 generated the same predictions as algorithm 2. Further lowering the coefficients of $a(n)$ is guaranteed to not affect the predictions for the eventual outcomes under CORSIA+ or the Simple MGF Mechanism. Various numerical simulations suggested that further lowering the coefficients of $a(n)$ does not affect the predictions for the MGF Mechanism with PMFs, either.

7.1 Overview of results for r ranging from 0 to 1 for each mechanism

I now provide some summaries of the results that were obtained for the case where the set of initial participants is $\{China, EU\}$. The motivation for considering this set of initial participants is as follows: The EU has been the leading actor in efforts to mitigate global aviation emissions, as documented in 3.1, and considers taking action in addition to CORSIA, as explained in 4. The EU could try to convince other countries to join in initiating the mechanisms. A plausible candidate country for joining in initiating the MGF Mechanisms might be China since these mechanisms would satisfy China's demand, articulated in the con-

text of CORSIA, that participating countries should be able to decide to which GPGI the carbon pricing revenue collected from its flights will be allocated.

Algorithm 3 was used to predict the eventual outcome of the game for each of the three mechanisms and for values of r ranging from 0 to 1 in steps of 0.05. Thus the strategy profile that algorithm 3 converges to is taken to be the prediction of the eventual outcome. For CORSIA+ this prediction is actually deterministic, so it suffices to run the algorithm once for each value of r . However, for the Simple MGF Mechanism and the MGF Mechanism with PMFs, algorithm 3 provides a probabilistic prediction. I used a sample size of 20 for each value of r . Using this sample, I estimate the following quantities, which will be displayed in the graphs below:

‘maximal aggregate payoffs’: For each run of algorithm 3, I take the eventual strategy profile that it converged to, compute the corresponding payoffs and sum them together. Then I take the maximal value amongst all runs of the algorithm.

‘expected aggregate payoffs’: For each run of algorithm 3, I take the eventual strategy profile that it converged to, compute the corresponding payoffs and sum them together. Then I average over all runs of the algorithm to estimate the expected value.

‘maximal aggregate revenue raised for GPGIs’: For each run of the algorithm 3, I take the eventual strategy profile that it converged to, compute the corresponding allocations to the GPGIs and sum them together. Then I take the maximal value amongst all runs of the algorithm.

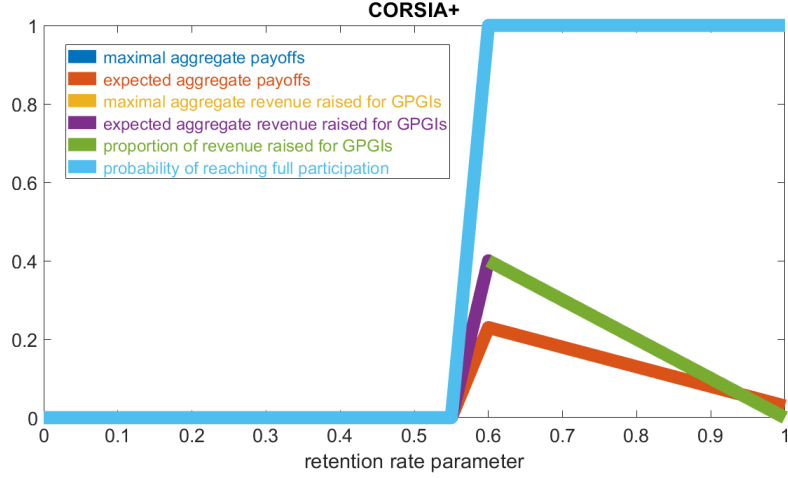
‘expected aggregate revenue raised for GPGIs’: For each run of the algorithm 3, I take the eventual strategy profile that it converged to, compute the corresponding allocations to the GPGIs and sum them together. Then I average over all runs of the algorithm.

‘proportion of revenue raised for GPGIs’: For each run of the algorithm 3, I take the eventual strategy profile that it converged to, compute the corresponding allocations to the GPGIs and sum them together. Then I average over all runs of the algorithm, thus obtaining the average money raised for GPGIs. Moreover, for each run of the algorithm, I take the eventual strategy profile that it converged to, compute the corresponding aggregate tax revenue raised. Then I average over all runs of the algorithm, thus obtaining the average tax revenue collected. Finally, I compute the ratio of the average money raised for the GPGIs over the average tax revenue collected.

‘probability of reaching full participation’: I estimate this by computing the proportion of runs of the algorithm where it converges to an eventual strategy profile at which all players participate.

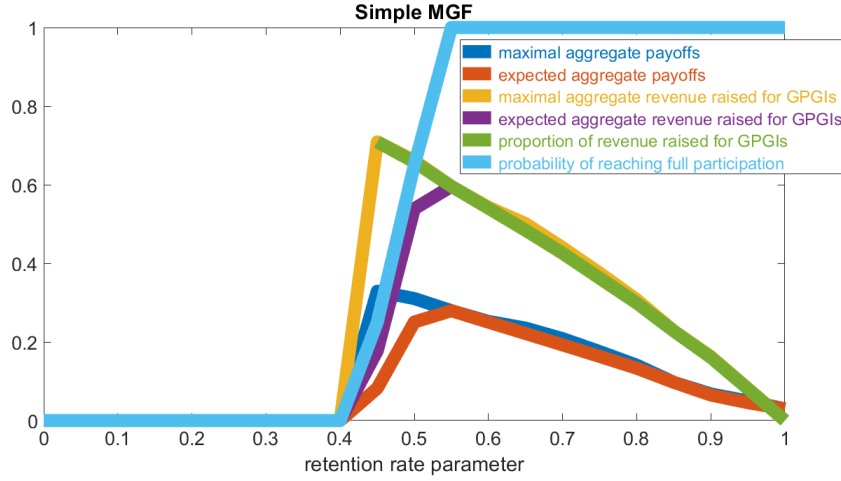
As a normalisation, I set to 1 the tax revenue that would be collected if all players participate.

7.1.1 CORSIA+



For $r \leq 0.55$ the algorithm 3 always converges to a Nash equilibrium with no participation. For $r \geq 0.6$ the algorithm always converges to a Nash equilibrium with full participation.

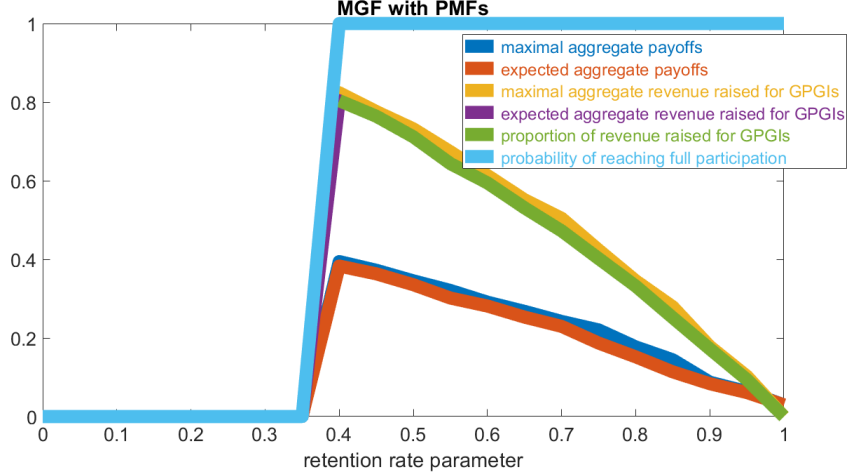
7.1.2 The Simple MGF Mechanism



For $r \leq 0.4$ the algorithm 3 converges to the Nash equilibrium with no participation in all runs for the Simple MGF Mechanism. At $r = 0.45$ it starts becoming possible for the algorithm 3 to converge to a strategy profile with full participation. In fact, it turns out that in almost all runs of the algorithm it

either converges to a Nash equilibrium with full participation or a Nash equilibrium with no participation. There is only one value of r , namely at $r = 0.5$, where sometimes the algorithm converges to a Nash equilibrium with partial participation.

7.1.3 The MGF Mechanism with Proportional Matching Funds (PMFs)



For $r \leq 0.35$ the algorithm converges to a Nash equilibrium with no participation in all runs for the MGF Mechanism with PMFs. For $r \geq 0.4$ the algorithm 3 always converges to an approximate Nash equilibrium with full participation.

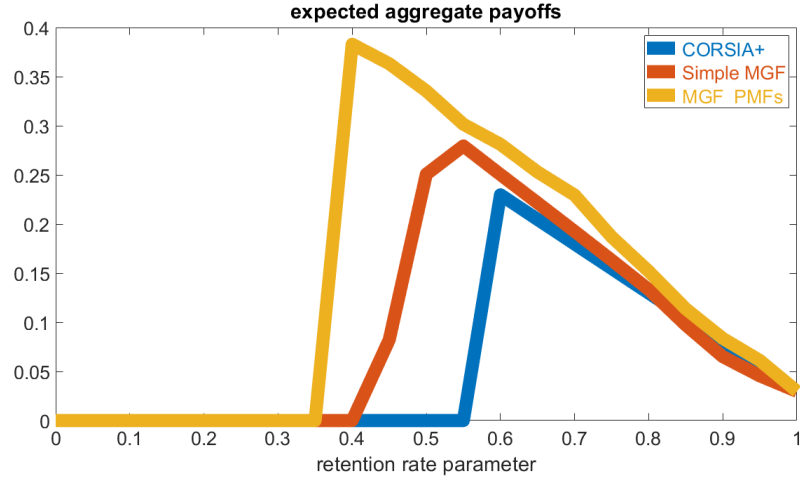
7.2 Direct comparison between the three mechanisms

The following graph shows a direct comparison of the expected aggregate payoffs generated by the mechanisms:

We see that in the simulation the MGF Mechanism with PMFs achieves the highest expected aggregate payoffs for all values of the retention rate parameter. For the MGF Mechanism with PMFs, the expected aggregate payoff is maximised at $r = 0.4$, where it is 0.38. For the Simple MGF Mechanism, the expected aggregate payoff is maximised at $r = 0.55$, where it is 0.28. For CORSIA+, the expected aggregate payoff is maximised at $r = 0.6$, where it is 0.23. Thus under the Simple MGF mechanism 1.22 times as much aggregate payoff can be achieved as under CORSIA+ and under the MGF Mechanism with PMFs 1.67 times as much aggregate payoff can be achieved as under CORSIA+.

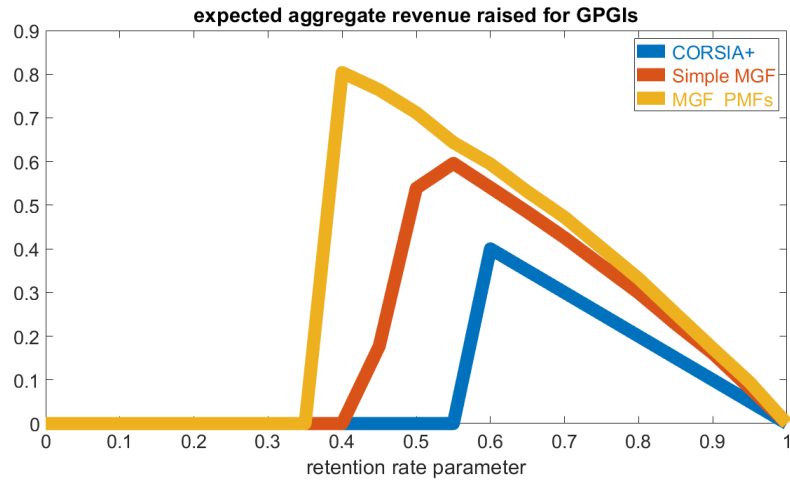
From the graph one can see that the main source of welfare gains generated by the mechanism is due to the benefits generated by the money raised for GPGIs rather than due to the emissions reductions induced by the taxes on the aviation emissions themselves. Indeed, for $r = 1$ a Nash equilibrium with full participation is always reached but no money is raised for GPGIs. The expected aggregate payoffs at this retention rate of $r = 1$ is 0.03.

Figure 4:



The following figure shows that the higher aggregate payoffs of the MGF Mechanism with PMFs is mostly explained by the fact that under it, a larger aggregate amount of money is raised for GPIs:

Figure 5:

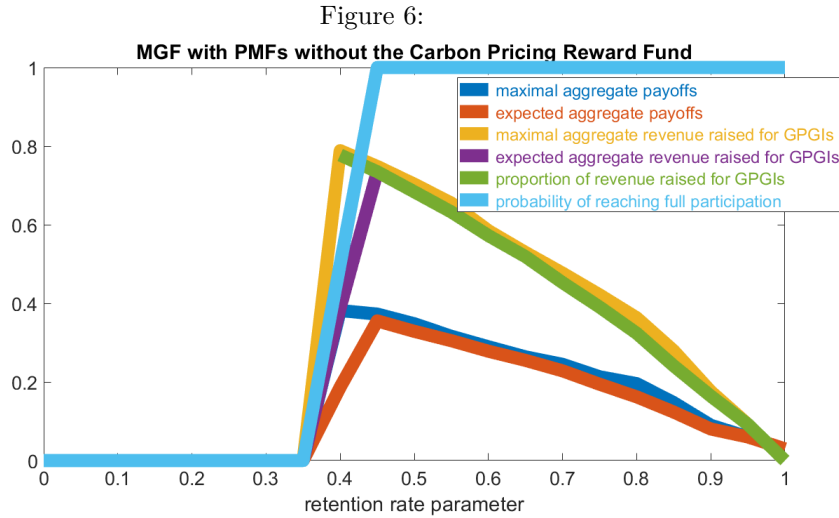


The maximal amount of expected revenue raised for GPIs is 0.40 for CORSIA+, 0.60 for the Simple MGF Mechanism and 0.80 for the MGF Mechanism with PMFs. The annual revenue of the aviation industry is \$855 billion as of 2018. Suppose the carbon tax x imposed in the treaty mechanisms on international aviation emissions is set to \$36, which is the social costs of carbon estimate from Nordhaus (2017). This would amount on average to a tax burden

of about 10% of current ticket prices. Suppose all countries participate. If one assumes that the total price elasticity¹⁹ of demand is -0.3, then one finds that \$83 billion would be raised per year if all flights were covered. However, all the mechanisms studied in this paper include only international flights²⁰, which contribute (the other 40% being due to domestic flights). Thus we obtain the prediction that the MGF Mechanism with PMFs with the emissions tax rate x set so as to amount on average to 10% of current ticket prices and with a retention rate of $r = 0.4$, would raise $0.60 * 8 * \$83 = \40 billion per year.

7.3 Comparative Statics with respect to the set S of eligible GPGIs

Rerunning the algorithm for the MGF Mechanism with PMFs for the case where the Carbon Pricing Reward Fund is removed from the set of eligible GPGIs, yields the following results:



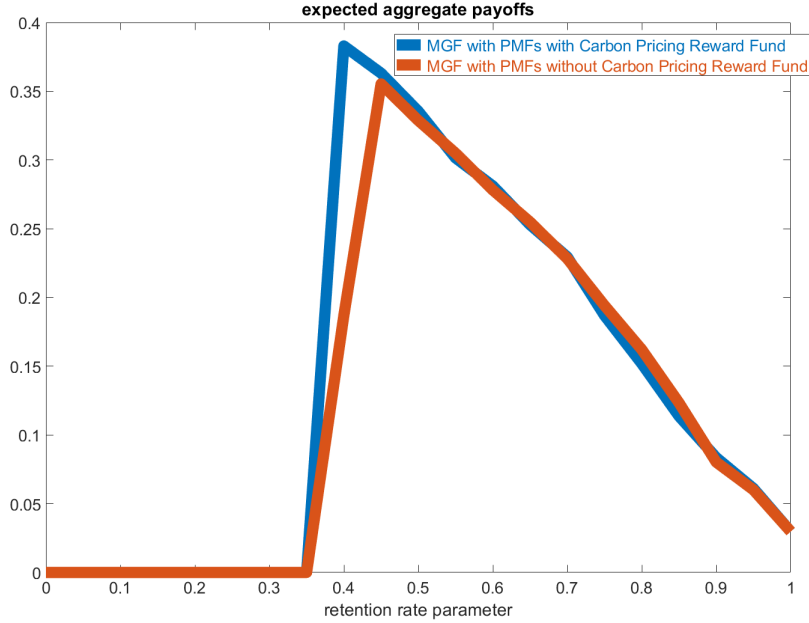
Whilst with the full set of the 6 GPGIs, including the Carbon Pricing Reward

¹⁹From IATA an estimate of the average price elasticity of demand is available, which is -0.6. This estimate is an average of elasticities for price changes within and between regions closely corresponding to the regional blocks that are the players in our model. Thus these estimates do not take into account substitution to flights to other regions. If one assumes that 50% of the reduction in demand caused by an increase in the tax on aviation emissions within a region or between two regions leads to substitution to flights to other regions, one obtains -0.3 as the estimate.

²⁰In line with the current CORSIA negotiations, I have focused in this article on mechanisms only concerning emissions from international aviation. However, it could be better to establish mechanisms that would also include taxation obligations on domestic flights, as Chancel and Piketty (2005) propose. I leave it for future work to repeat the analysis presented here for a modified version of the mechanisms, in which the taxation obligations would include domestic flights.

Fund, the algorithm converges for $r = 0.4$ for all runs to an approximate Nash equilibrium with full participation, we see here that this no longer is the case, once the Carbon Pricing Reward Fund is removed. This affects the expected aggregate welfare:

Figure 7:



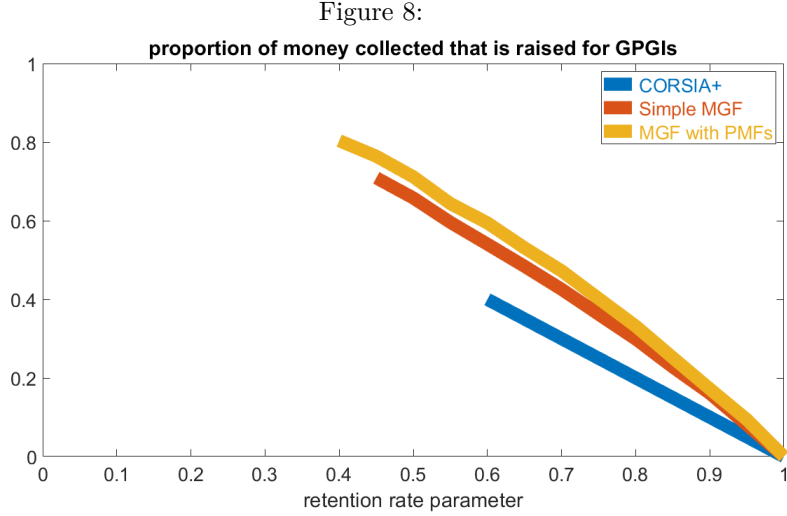
Removing the Carbon Pricing Reward Fund from the set of eligible GPGIs reduces the achievable welfare gains by 7%.

7.4 Legitimacy and the risk of the mechanisms triggering trade wars

Throughout the analysis, we have assumed away the possibility that non-participating countries might oppose the mechanisms and retaliate against the participating countries. The EU's experience from 2012 that we outline in 3.1 shows that this danger needs to be taken very seriously. In 2012 many observers were concerned that the policy might trigger a trade war. For evaluating which mechanism should best be initiated, we need to take into account how each possible mechanism will affect the probability of a trade war.

It seems plausible that the more a mechanism contributes to the welfare of non-participants, the less likely it is that they will oppose it. Firstly, this is because they have less material incentive to try to cause the mechanism to disappear. Secondly, the mechanism will presumably be more legitimate if the

participating countries allocate most of the money to GPGIs rather than keeping it for themselves (as the EU did in 2012). The MGF mechanism with PMFs again performs best in this regard:



There is another reason why the MGF mechanisms could have more legitimacy than CORSIA+: Allowing participating countries to decide how to allocate to GPGIs the money that they collect as taxes would accommodate the demands of several countries, including Brazil and China, who reportedly in the context of the CORSIA negotiations reject the idea that the list of eligible offset standards should be determined by the ICAO Council and prefer that States be allowed to apply their own offset standards. According to Andrew Murphy from the NGO Transport and Environment, who is an observer in the CORSIA negotiations: “The reason for China’s scepticism is that, as a point of principal, they don’t want to be told by a UN agency which offsets they have to use for their airlines.”²¹

These considerations thus all point in the direction of the MGF mechanisms having more legitimacy and incurring a lower probability of triggering a trade war than CORSIA+.

8 Conclusion and discussion

In this study, I have defined and analyzed two new proposals for club mechanisms for financing multiple Global Public Good Institutions (GPGIs) via taxes on international aviation emissions. Participating countries would be obliged to tax at a fixed rate all the emissions from outgoing flights and also from all flights

²¹This position can explain why, as of the time of writing (september 2019), China has not decided to participate in the voluntary first phase of CORSIA.

coming in from non-participants. In the first new proposal, called the Simple MGF Mechanism, participating countries could allocate the tax revenue that they collect to any GPGIs defined as eligible in the treaty. Of the tax revenue that they decide to not directly allocate to GPGIs, they could retain a fixed proportion r , where the retention rate parameter r would be fixed in the treaty. The remaining proportion, $1 - r$, of the money not directly allocated would be added to the GPGIs in proportion to the direct allocations made to them, thus enhancing the influence of those that make direct allocations to GPGIs.

The second proposal, called the MGF mechanism with Proportional Matching Funds, enriches this mechanism by allowing countries to also allocate the tax revenue to so-called Proportional Matching Funds, that support some subset of eligible GPGIs by distributing their money to them in proportion to the direct allocations made to them.

I compare these new proposals to an alternative mechanism, called CORSIA+, that is based on current proposals and that would fund only a single GPGI. To do this, I attempt to predict what would happen if the EU and China were to initiate one of these three mechanisms, using an iterated best response algorithm and using empirical estimates for the payoffs that countries would derive from the different GPGIs and as a result of the aviation emissions taxes themselves. In the simulations, I find that the MGF Mechanism with Proportional Matching Funds generates the largest expected aggregate welfare for each value of the retention rate parameter. If one chooses for each mechanism the retention rate parameter so as to maximise the expected aggregate welfare that it generates, then one finds that the Simple MGF Mechanism generates 1.22 times as much welfare and raises 1.5 times as much aggregate revenues for the GPGIs. The MGF Mechanism with PMFs generates 1.67 times as much welfare as CORSIA+ and raises 2.0 times as much aggregate revenue for the GPGIs. For a carbon tax of \$36, which would increase ticket prices by about 10%, the MGF Mechanism with PMFs would raise about \$40 billion for GPGIs.

One questionable assumption is that in the context of the mechanisms, the countries within each of the 12 regional blocks overcome their collective action problems. This seems at first glance very implausible. However, in the context of the MGF mechanism with Proportional Matching Funds, the gains from cooperation within each block might be particularly large, which presumably would make cooperation within each block more likely.²²

Another questionable assumption is the absence of any foresight that has been implicit in the iterated best response algorithm that we have used. In all of the mechanisms that we have studied here, the following holds: by participating, one decreases the incremental cost for the others to join. Thus foresight would

²²To see this, consider for example the gains from cooperation for African countries that arise endogenously at the Nash equilibria described in section 6.4.3: All African countries most benefit from money going to the Global Fund Against AIDS, TB and Malaria. By participating and giving money to this GPGI, African countries can cause not just the tax money that they collect to go to this GPGI but also additional money that comes from the PMF with Global Fund Against AIDS, TB and Malaria, CEPI and Forest Carbon Partnership's Carbon Fund, to which the Middle East allocates all its tax revenue.

weigh in favor of joining since by doing so one can cause joining to be more attractive (and thus more likely) for others. Taking into account this effect would presumably improve the performance of all the three mechanisms to a similar extent. However, there is a further effect specific to the MGF with PMFs: By giving to a PMF supporting a set J of GPGIs, one can increase the incentives to join for others who strongly prefer some GPGI from J over the other GPGIs in J . This suggests that if one were to assume that players reason with foresight, then the performance of the MFG mechanism with PMFs would be improved by more than that of the other mechanisms. Future modelling work could investigate whether the conclusions of this line of informal reasoning do in fact hold.

An important further research question is how the set S of eligible GPGIs will affect the performance of the MGF mechanisms. It is plausible that by adding further GPGIs which benefit different players, the performance of the MGF mechanisms will typically be improved. An example of a GPGI that could be added to the set S is a global fund that would buy up fossil fuel deposits that are marginally profitable, as suggested by Harstad (2012). This would benefit the countries possessing such deposits, as they will earn rents from the sale of the deposits. Another GPGI to add to the set S could be a fund that would reward countries for funding research on clean energy. Such a fund could build on the Mission Innovation Initiative that was established in 2015. A better understanding of the effect of the set S on the performance of the mechanisms could be useful for informing the initial choice for the set S when the mechanism is started. Moreover, it could also help in the design of mechanisms for updating the set S over time.

Appendix

A Matlab programs

The Matlab code with which all the simulations were computed is available here: <https://github.com/gpgi-financing/mgf-mechanisms>. Running the script “main.m” will redo all the simulations and also produce all the graphs displayed in this paper, with a run time of about 10 hours. In section 1.1 of the script, all the parameters are listed for which I guessed the values. The reader is encouraged to modify these parameters and rerun the simulations. The reader can also send me the profiles of these parameter values that he finds most plausible. I will run the simulations for these parameter values and report in <https://github.com/gpgi-financing/mgf-mechanisms/tree/master/figures> the results, as well as any explanations for the guesses of parameter values that the corresponding reader wants to provide. I will also report all such simulation results and explanations on <https://sites.google.com/view/mgf-mechanisms/accueil>.

B Estimation of the payoffs

Now we will compute values for the various parameters, using econometric estimates and data when possible, whilst resorting to ad hoc numerical assumptions for several of the parameters.

Let $K = \{\text{Africa, China, EU, Eurasia, India, Japan, Latin America, Middle East, Other high-income countries, Russia, US, other Asia}\}$ denote the set of players. For $k \in K$ let U_k denote player k 's payoff. Since we are assuming that each player's payoff is the sum of the payoffs of the countries that it comprises, we have:

$$U_k = \sum_{i \in k} u_i = \sum_{i \in k} t_i - c_i x F_i + Y_i + b_i x G + \sum_{s \in S} \theta_i^s Z_s \quad (4)$$

Throughout I will assume that in the games induced by the mechanisms countries coalesce into the 12 blocks given by the set K , each one of them acting as a unitary player whose payoff is the sum of the payoffs of the countries that constitute it.

B.1 The tax burden coefficients c_i and the redistributive effects Y_i

For simplicity, we shall assume throughout that the tax on the carbon emissions is equivalent to an ad valorem tax and that there are only scale effects. In other words, we neglect the fact that airlines will reduce emissions not just by decreasing the number of flights but also through other ways such as by increasing load factors.

We will assume that the tax rate x is fixed and small. Let us consider $-c_i x F_i + Y_i$. Plugging in our linear separability assumptions, i.e. that $c_i = \sigma_i + \eta_i$ and that $Y_i = \sum_{j \neq i, k \neq i: j \text{ or } k \text{ participate}} \gamma_{ijk} x f_{jk}$ we obtain:

$$-\sum_{i \in k} c_i x F_i + Y_i = -\sum_{i \in k} (\sigma_i + \eta_i) x F_i + \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate}} \gamma_{ijl} x f_{jl} \quad (5)$$

Recall from section 5 that η_i and γ_{ijk} corresponds to redistributive effects such as the shifting of rents from tourism across countries and terms of trade effects. We shall assume that these effects are zero-sum for the world as a whole. This means that $\sum_k (-\sum_{i \in k} \eta_i + \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate}} \gamma_{ijl} x f_{jl}) = 0$. In particular, to compute aggregate welfare we can simply ignore the η_i and the γ_{ijk} .

We shall assume throughout that each country internalises only the welfare of its own citizens. Thus when we sum the welfare effects $\sum_{i \in I} \sigma_i x F_i$ over the set of all countries, each consumer is counted exactly once. We shall assume that for global welfare excluding climate change externalities, the marginal tax

burden from international aviation is equal to that of other taxes²³. More precisely, we shall assume that this marginal tax burden from international aviation (excluding externalities) is 1²⁴, which means that $\sum_{i \in I} \sigma_i x F_i = \frac{1}{2} \sum_{i \in I} x F_i$. For simplicity and transparency, we shall assume that the σ_i are all equal, which implies that they are all equal to 0.5.

Let us assume that at the status quo each country has optimally (from its own perspective) chosen its level of taxation on international aviation, assuming that it can retain the tax revenue raised from it²⁵. This implies that $\sigma_i + \eta_i = 1 \forall i$. Given that $\sigma_i = 0.5 \forall i$, we thus obtain that $\eta_i = 0.5 \forall i$.

Let us rearrange $-\sum_{i \in k} (\sigma_i + \eta_i) x F_i + \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate}} \gamma_{ijl} x f_{jl}$ to separate out the part that is affected by whether or not player k participates:

$(-\sum_{i \in k} (\sigma_i + \eta_i) x F_i + \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate and } j \in k \text{ or } l \in k} \gamma_{ijl} x f_{jl}) + \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate, } j \notin k, l \notin k} \gamma_{ijl} x f_{jl}$
Thus $\sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate and } j \in k \text{ or } l \in k} \gamma_{ijl} x f_{jl}$ is the redistributive effect that the countries making up player k will internalise thanks to the fact that they have coalesced into a single player in the context of the mechanisms.

Now we shall make the ad hoc assumption that each of the players that consist of several countries internalises half of the redistributive effects. As a result, we can now write:

$-\sum_{i \in k} c_i x F_i + Y_i = -\tilde{c}_k x \sum_{i \in k} F_i + \sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate, } j \notin k, l \notin k} \gamma_{ijl} x f_{jl}$
where $\tilde{c}_k = 1$ if k consist of a single country (which is the case for k =China, India, Japan, Russia, US) and $\tilde{c}_k = 0.75$ if k consists of several countries (which is the case for k =Africa, EU, Eurasia, Latin America, Middle East, Other high income countries, other Asia). Moreover, only $-\tilde{c}_k x \sum_{i \in k} F_i$ but not $\sum_{j \neq i, l \neq i: j \text{ or } l \text{ participate, } j \notin k, l \notin k} \gamma_{ijl} x f_{jl}$ is affected by the choice of player k .

B.2 The flight matrix f_{ij}

To estimate the network of emissions from international flights we use World-Pop's open-source global flight data. From this data, we computed for each pair of airports A and B the available seat kilometers by multiplying the estimated number of seats flying directly from A to B by the spherical distance computed from the longitudes and latitudes. These numbers were taken for the

²³The Atkinson Stiglitz theorem (1976) would predict that the marginal tax burden (excluding externalities) from taxes on international aviation is less than that of other taxes, since international aviation is currently taxed much less than other commodities. However, we assume equality for simplicity and transparency.

²⁴see Jacobs (2018) for a justification

²⁵The current profile of ticket taxes could indeed be interpreted as a Nash equilibrium in a game where each country acts individually. Countries do not coordinate on ticket taxes. Moreover, by now it appears to be accepted that it is legal for each country to set ticket taxes (Faber and Huigen 2018). Hence each individual country's choice of ticket tax can be seen as a best response.

However, taxes on emissions are likely to have a lower σ_i (see Stern (forthcoming) for a theoretical analysis suggesting that switching from ticket taxes to emissions taxes will increase welfare by increasing load factors). The current absence of emissions taxes is the result of legal obstacles mentioned in (Larrson et al 2018). We leave it for future work to refine the model by taking these effects into account.

simulations as an estimate for the associated greenhouse gas emissions.²⁶

B.3 The global effects $b_i(x)$ from the emissions taxes on flights

As stated in the previous section, we assume that the taxes on aviation emissions are equivalent to an ad valorem tax and that there are only scale effects. In other words, we neglect the fact that airlines will reduce emissions not just by decreasing the number of flights but also through other ways such as by increasing load factors.

As mentioned before, we take into account two global effects that are caused by the reduction in oil consumption induced by a tax on aviation emissions: a reduction in climate change damages and a reduction in the oil price. We shall assume that the supply of flights is infinitely price elastic. The elasticity of aviation demand is denoted by ϵ_D .

First, let us compute the reduction in climate change damages. For this, consider a carbon tax equal to 1% of ticket prices. Such a tax will reduce aviation emissions by ϵ_D percent. The total annual amount of emissions from aviation are 859 million tonnes of CO₂. In line with Nordhaus (2017) we assume a social cost of carbon of \$36²⁷. The IPCC estimates that the climate forcing effect of aviation emissions is 3 times that of its carbon content, due to the high altitudes at which they take place. Based on this, we assume that the social cost of aviation is 3 times that of its CO₂ content contribution. Thus the total social cost of carbon from aviation is $3 \cdot \$36 \cdot 859 \cdot 10^6$.

The carbon tax of 1% of ticket prices thus averts $3 \cdot \$36 \cdot 859 \cdot 10^6 \cdot 1\% \cdot \epsilon_D$ of net climate damages due to aviation emissions. However, there is a rebound effect due to induced decrease in the price of oil, which according to Stoft (2008) is 26%. We must therefore multiply by $1 - \frac{1}{3} \cdot 0.26$. Since the annual revenue of the aviation industry is $\$855 \cdot 10^9$, the revenue generated by the carbon tax of 1% of ticket prices is approximately $\$855 \cdot 10^9 \cdot 1\%$. Thus we obtain that per dollar of carbon tax collected the averted climate change damages are equal to:

$$\text{aggregate averted climate change damage} = (1 - \frac{1}{3} \cdot 0.26) \frac{3 \cdot \$36 \cdot 859 \cdot 10^6 \cdot \epsilon_D}{\$855 \cdot 10^9} = 0.108 \cdot \epsilon_D$$

From IATA an estimate of the average price elasticity of demand is available, which is 0.6. This estimate is an average of elasticities for price changes within and between regions closely corresponding to the regional blocks that are the players in our model. Thus these estimates do not take into account substitution to flights to other regions. We shall assume that 50% of the reduction in demand caused by an increase in the tax on aviation emissions within a region or between

²⁶In december 2019, all participants to CORSIA will report emissions data.

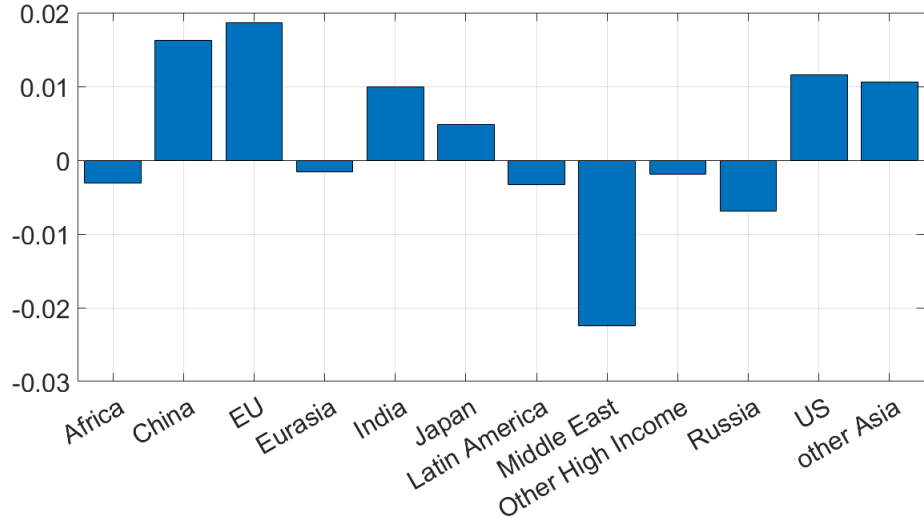
²⁷The relevant concept of social cost of carbon for our purposes is descriptive: For predicting the evolution of the game, as we will try to do in 7, we need to estimate how governments will in fact evaluate the benefits of climate change mitigation. Whilst the normative arguments against pure time discounting are compelling, it seems more plausible to model countries as agents that use pure time discounting, as done in the RICE model, on which the \$36 SCC estimate is based.

two regions leads to substitution to flights to other regions. Thus we need to multiply our estimate by 0.5, giving us $\epsilon_D = 0.5 \cdot 0.6 = 0.3$. With this we obtain *aggregate averted climate change damage* = 0.0297.

Now let us compute the redistributive effect that happens due to the change in oil prices on the world market. From the above calculation, we already have an estimate for the amount of averted CO2 emissions: $\frac{\text{averted tons of CO}_2}{\$ \text{ of tax revenue collected}} = (1 - 0.26) \frac{859 \cdot 10^6 \cdot \epsilon_D}{\$855 \cdot 10^9}$. Now we need to estimate $\frac{\text{loss to oil producers}}{\text{averted tons of CO}_2} = \frac{\text{loss to oil producers}}{\text{barrels of crude oil}} \frac{\text{barrels of crude oil}}{\text{averted tons of CO}_2} = \frac{dP \cdot Q}{dQ} \frac{\text{barrels of crude oil}}{\text{averted tons of CO}_2} = P \frac{\frac{dP}{P}}{\frac{dQ}{Q}} \frac{\text{barrels of crude oil}}{\text{averted tons of CO}_2}$. Stoft (2008) calls $\frac{\frac{dP}{P}}{\frac{dQ}{Q}}$ the “oil change ratio” and gives 5 estimates for from the literature, ranging from 1.5 to 12. We assume the lowest value, 1.5, (as Stoft does) so as not to risk exaggerating the redistributive effects which turn out to help the performance of the MGF mechanisms. For the oil price P we take the average price during 2018. $\frac{\text{barrels of crude oil}}{\text{averted tons of CO}_2}$ has been estimated to be 3.15. Putting all these estimates together, we obtain:

$$\frac{\text{losses for crude oil producers}}{\$ \text{ of tax revenue collected}} = 0.068$$

These marginal losses for crude oil producers are of course equal to the marginal gains for oil consumers. To compute the aggregate redistributive effect for each of the 12 players, we use data for crude oil production and oil consumption from Enerdata. By adding these results with the estimates for the effects due to reduced climate change we obtain the global effects $\mathbf{b}_i(\mathbf{x})$ from the emissions taxes on flights, displayed in the following diagram:



B.4 Effects of the money raised for the GPGIs: θ_s^i

I start by deriving an estimate for the aggregate payoffs generated per dollar given to the Forest Carbon Partnership’s Carbon Fund, i.e. for $\sum_{i \in I} \theta_s^i$, where I denotes the set of all countries. I then explain the assumptions made for the

aggregate payoffs for the other GPGIs. Finally, I will detail how the payoffs for all the players are computed for each GPGI.

The Forest Carbon Partnership’s Carbon Fund rewards governments for reducing deforestation. Suppose a dollar is added to its budget. As a result, reward payments are increased. This has two effects: Firstly, the governments’ rents are increased and secondly, the governments will reduce deforestation. Suppose that half of the additional dollar accrues as additional rents to governments. We now need to know: By how many tons of CO₂ will emissions be reduced? <https://www.nature.com/articles/s41558-019-0485-x> provide the most recent estimates for what the marginal abatement cost curves will be for reduced deforestation for the period 2021-2031. However, to know the marginal abatement costs, we need to make an assumption of how much overall funding will be provided for reducing deforestation. Donor contributions have been about a billion per year during the last years. However, countries have also pledged to reduce deforestation as part of their nationally determined contributions under the Paris agreement. Piris-Cabezas et al. (2018) estimates that in an “intermediate ambition scenario” the marginal abatement cost in 2020 will be \$19.4. Combining this with our assumption that the social cost of carbon is \$36 (from Nordhaus (2017)), one finds that the marginal global benefit-cost ratio of reducing deforestation is about 2. Since we are assuming that half of each dollar given to the Forest Carbon Partnership accrues as rents to governments, we can deduce that per dollar given to the Forest Carbon Partnership we have: $aggregate\ payoffs = 0.5 + 0.5 * 2 = 1.5$.

I have not found studies that would allow me to do an analogous calculation to compute the aggregate payoffs for the other GPGIs. 3 of the other 5 GPGIs are similar to the Forest Carbon Partnership’s Carbon Fund in that they already exist and contribute to a global public good whose benefits are widely spread around the world: The Fund buying up CDM certificates and ITER contribute to climate change mitigation and CEPI to preventing global pandemics. For transparency, I will assume that these GPGIs’ aggregate payoffs per dollar used are also 1.5. The Carbon Pricing Reward Fund would also contribute to climate change mitigation. However, to be able to rationalise the fact that this GPGI does not yet exist, I assume that the net benefit created per dollar used is 20% lower, i.e. I assume that the aggregate payoffs that it creates per dollar received is only 1.4. The Global Fund for AIDS, TB and Malaria is different in that the benefits that it creates are mostly concentrated in the countries with a high prevalence of the diseases it addresses. This motivates why I use a different method for estimating the payoff it creates, reported in B.4.6.

As explained in 7, the GPGI assumed to be selected under CORSIA+ forwards its money in equal proportions to the Fund buying up CDM Certificates and the Forest Carbon Partnership’s Carbon Fund. From this it follows that the aggregate payoffs that it generates are $\frac{1.5+1.5}{2} = 1.5$.

To summarise, here is an overview of the aggregate payoffs:

Global Public Good Institution	Aggregate Payoffs
Fund buying up CDM Certificates	1.5
CEPI (Coalition for Epidemic Preparedness innovation)	1.5
Global Fund Against AIDS, TB and Malaria	1
Forest Carbon Partnership’s Carbon Fund	1.5
ITER	1.5
Carbon Pricing Reward Fund	1.4
GPGI assumed to be selected under CORSIA+	1.5

We note that these aggregate payoffs are quite favorable to CORSIA+. However, we will find that the MGF Mechanisms will outperform CORSIA+.

A further parameter that we do not derive from data but instead set based on plausibility considerations, is the proportion of money that each GPGI ends up paying out as rents, as we will detail in the explanations below.

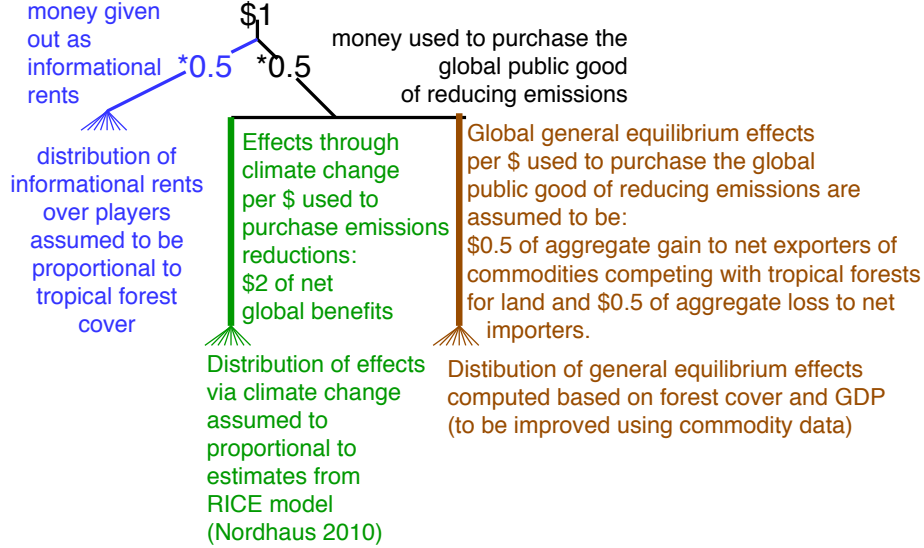
B.4.1 Effects per dollar given to the Forest Carbon Partnership’s Carbon Fund

The Forest Carbon Partnership rewards countries with tropical forests for conserving them. An elaborate methodology has been developed through which reference levels of deforestation are determined. Reward payments are made for reductions in deforestation below the reference levels. If the reference level for a country is set too high, then the country will not respond to the incentives provided. If the reference level is too low, it will dilute the incentives created, since most of the reward payments will pay for efforts that the country would do anyway. Since the countries’ costs of reducing deforestation are heterogeneous and their private information, most participating countries will earn information rents.

Suppose, hypothetically, that the Forest Carbon Partnership had complete information about the countries’ payoffs. Suppose it wants to achieve a certain profile of results (i.e. an amount of deforestation reduction for each country with tropical forests). Then it could make each country a take-it-or-leave-it offer whereby it would receive a certain reward payment iff it achieves the specified results. To achieve the specified result with the lowest possible payments, the Forest Carbon Partnership could tailor the reward payment so as to make the country indifferent between accepting the offer and ignoring it. In reality, the Forest Carbon Partnership has only incomplete information about the countries’ costs and will therefore inevitably end up paying most countries more than necessary to induce them to respond to the mechanism rather than to ignore it.

However, it is useful to decompose any mechanism contracting under incomplete information as the sum of the mechanism contracting under complete information and achieving the same profile of results without leaving any rents to the participants and a mechanism that consists of only paying out rents. In other words, we can view any actual mechanism as if it first “purchased the global public good of reducing emissions by contracting under complete information and leaving no rents to the countries” and then “paying out rents”.

Again, we can think of an additional dollar as doing a combination two things: Firstly, paying out information rents and secondly, purchasing the global public good (in this case reducing emission due to tropical deforestation) by compensating the agents in the mechanism (in this case national and local governments) for the costs incurred to increase the provision of the global public good.



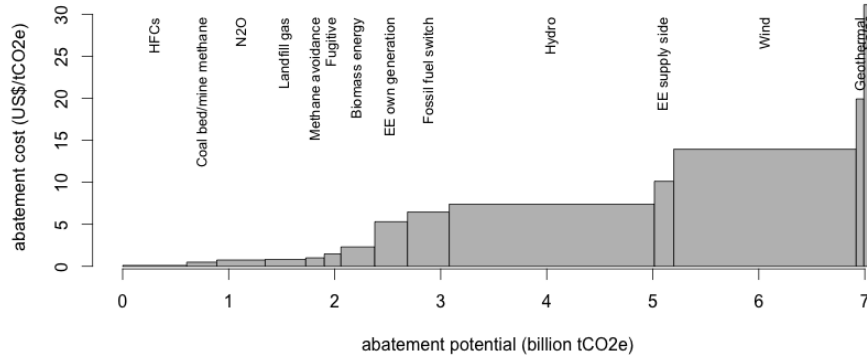
As explained above, we assume that per dollar given to this GPPI, 0.5 accrue as rents to the countries receiving reward payments. Per dollar used to purchase the global public good of reducing emissions, 2 dollars of aggregate benefits are generated globally via averted climate change.

There is also a potentially important redistributive general equilibrium effect that we need to take into account: Some of the forest conservation measures that will be taken in response to the incentive payments provided by the Forest Carbon Partnership's Carbon Fund will entail reducing the production of commodities such as palm oil, pulp, paper, timber and soybeans that are produced by the countries with tropical forests. As a result, the world market price of these commodities increases, which benefits the sellers of these commodities.²⁸ The flip side of this is of course that the buyers of these commodities lose out. We make the ad hoc assumption that per dollar used to purchase emissions reductions, 0.5 dollars of benefits accrue to the sellers. We assume that the consumption of these commodities is proportional to GDP and that the production is proportional to the tropical forest cover. We leave it for future work to generate proper estimates using data on all the commodities that drive deforestation.

²⁸Harstad and Mideksa (2017) point to a countervailing effect that also needs to be taken into account: Higher prices for the commodities driving deforestation makes illegal logging more profitable, thereby making enforcement of conservation more costly.

B.4.2 Effects per dollar given to the Fund buying up CDM certificates

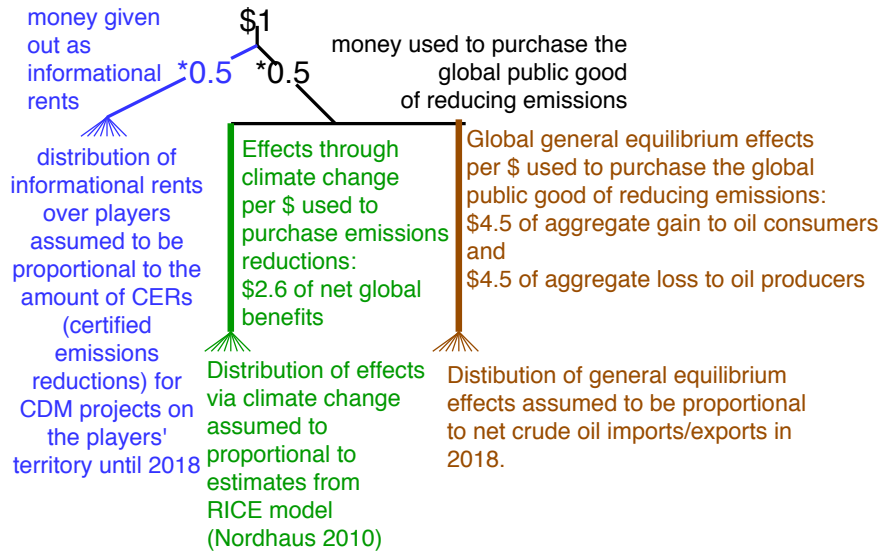
Consider the marginal abatement cost curve for registered CDM projects (from Shishlov (2014)):



We can think of the Clean Development Mechanism as if it was a principal contracting with project owners who are the agents in the mechanism. The specific mechanism implemented by the CDM is to pay project owners an amount proportional to the abated emissions. In practice, the payment per abated emissions is of course the market price of carbon offsets, which fluctuates over time. However, we shall abstract away from this feature and pretend that everything is static.

The abatement costs of any individual project is private information to the project owner who will earn information rents. If the principal had perfect information, he could cause the same projects to happen as he does under the actual mechanism, whilst not leaving any rents to the agents. This he could hypothetically do by making each agent a take-it-or-leave-it offer for doing the project, offering the project owner just enough money to make him indifferent between accepting the offer or not. The actual mechanism can now be viewed as a combination of this hypothetical contracting under perfect information and paying out gifts equal to the information rents in the actual mechanism. We can view the first part in this combination as “purchasing the global public good of reducing emissions” and the second part as “giving out information rents”.

The CDM has been criticised for having large rents. However, increasingly, low-cost abatement categories are covered under other mechanisms and are thus no longer eligible for new CDM projects. For example, HFCs are covered under the 2016 Kigali amendment to the Montreal protocol. Hence information rents are getting smaller. This motivates the following (admittedly ad hoc) assumption: We assume that in the actual CDM 50% of the money ends up being used for purchasing the global public good of reducing emissions. This is illustrated in the following diagram:



We assume that the players (i.e. Africa, China, EU, Eurasia, India, Japan, Latin America, Middle East, Other high-income countries, Russia, other Asia) internalise the rents accruing to the owners of projects on their territory. Moreover, we assume that the distribution of rents is proportional to the amount of CERs (certified emissions reductions) for CDM projects on the players' territories. We take the latest data found for this (counting all CERs until 2018)

Now let us turn to the effects of the actual emissions reductions. The players' payoffs are assumed to be affected in two ways. Firstly, the emissions reductions avert some climate change. We shall now make the ad hoc assumption that per dollar used to purchase emissions reductions 2.5 dollars of net global benefits arise²⁹. We assume that the distribution of these effects is proportional to the estimates from the RICE model (from Nordhaus 2010, also listed on page 14 of Nordhaus (2015)).³⁰

²⁹This number is broadly consistent with our assumption that the social cost of carbon is \$36 per ton of CO₂. To see this, suppose that the marginal abatement cost of the projects funded via the CDM will be \$15, in line with the marginal abatement cost curve shown above. This would imply that the benefit cost ratio would be 3.6. However, Cames et al (2016) find that "73% of the potential 2013-2020 Certified Emissions Reduction (CER) supply have a low likelihood that emission reductions are additional and are not over-estimated", so so we should substantially revise our number downwards from 3.6. This makes 2 seem reasonable.

³⁰Actually, to compute the overall payoffs, we make a slight adjustment because of the functioning of the clean development mechanism. According to the CDM guidelines projects are only eligible for receiving emissions reductions certificates if they can prove that they would not be profitable without the money from the sale of these certificates. One can show that if the marginal abatement cost curve is linear then the CDM does not change the incentives for the national government setting its subsidy for renewables.

The marginal abatement cost curve reported above already takes into account any national subsidies. The effective subsidy that project owners face is the sum of the national subsidy and the market price of the certificates for certified emissions reductions. We assume that the national subsidies are set individually optimally by each government. Therefore it is equal to

The second effect arising from emissions reductions is a redistribution from net oil exporters to net oil importers. From B.3 we know what this effect would be if all the emissions reductions were realised through reductions in oil consumption. Based on the data provided in <http://www.cdmpipeline.org/cdm-projects-type.htm>, I make the preliminary guess that the proportion of the CDM's emissions reductions that are through reduced oil demand is 0.1.

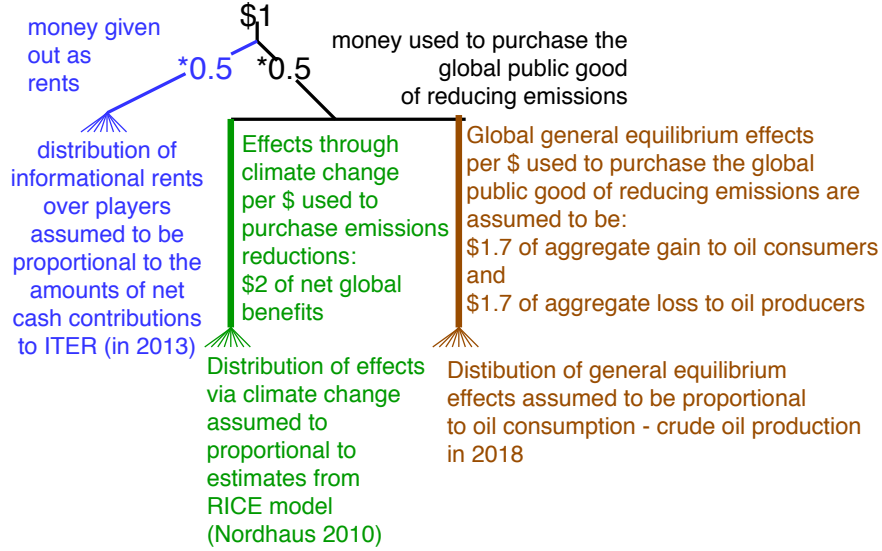
B.4.3 Effects per dollar given to ITER (The International Thermonuclear Reactor)

ITER has so far been financed through voluntary contributions by European Union, India, Japan, China, Russia, South Korea, and the United States. These countries will own the intellectual property that is generated, thus conferring rents to them. Whenever additional money is added to ITER (and assuming that the rules for access to the technology are not changed), the expected value of these rents is increased. Specifically, we will assume that the expected value of rents is increased by \$0.5 for every \$1 added to ITER. Analogously to our analysis of the CDM, we can think of a dollar reaching ITER to be split: \$0.5 are given out as rents and the remaining \$0.5 are “used to purchase the global public good of reducing emissions”. We assume that the rents accruing to the participating countries are proportional to the amounts of net cash contributions that they make. As far as the expected emissions reductions are concerned, the situation is assumed to be identical to the CDM.

Lastly, we need to make an assumption about how much the emissions reduction decrease the revenues for oil producers. The emissions reductions that ITER could cause would happen in several decades, if and when nuclear fusion reactors get deployed. Since the players plausibly discount the future at a large rate, these redistributive effects weigh less heavily in the players' payoffs than the immediate redistributive effects that the CDM induces. Based on this we make the ad hoc assumption that per averted emission ITER causes half as much redistributive effects via the oil price as the CDM.

the local externalities plus the global effects (see section B.3) that the country internalises. From this it follows that an additional project that happens due to the CDM causes not only the global benefits but also causes costs equal to the subsidy in place in the country where the project takes place.

In other words, since we assume that countries already set their national subsidies optimally from their perspectives, the global net benefit that an additional projects causes is computed by summing the benefits accruing to all the other countries but not to the country where the project takes place. Otherwise we would be double-counting.



B.4.4 Effects per dollar given to the Carbon Pricing Reward Fund

In contrast to the other 5 GPGIs that we consider, the Carbon Pricing Reward Fund does not yet exist. Cramton Stoft (2011) propose to create a global fund that would reward countries based both on how much they emit and how much net carbon tax revenue (i.e. carbon tax revenue minus fossil fuel subsidies) they collect (all sectors of the economy included). We shall consider the following version of this kind of proposal³¹:

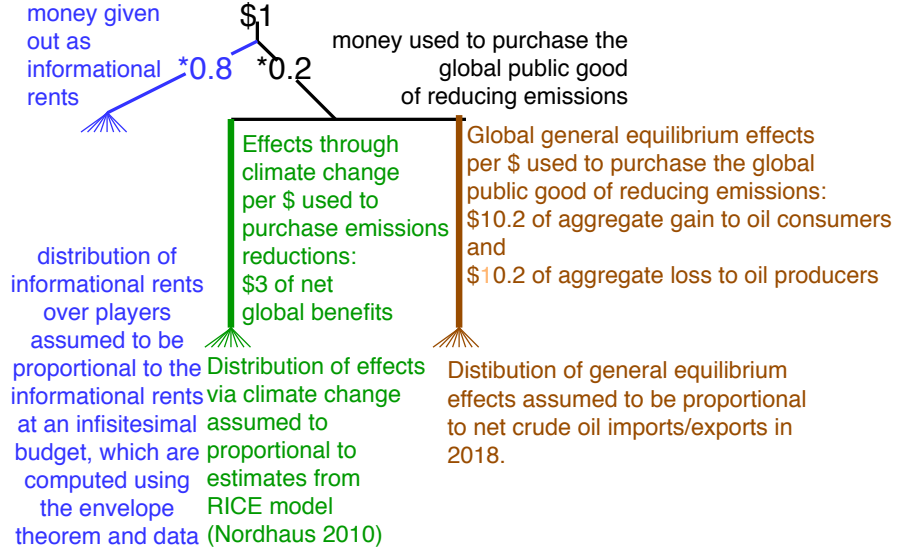
Definition: **The multiplicative mechanism** makes reward payments to countries based on their per capita consumption-based emissions Q , their net per capita carbon tax revenue R and their population P according to the formula:

$$f * \max(0, \hat{Q} - Q) * \max(0, R - \hat{R})P$$

where $\hat{Q}$ and $\hat{R}$ are fixed constants and f is endogenously adjusted so that the fund exactly uses up its budget.

We will assume that the Carbon Pricing Reward Fund uses the multiplicative mechanism. Given that the policies that the Carbon Pricing Reward would incentivise are very cost-effective, the question arises as to why such a mechanism does not already exist. One explanation could be that a large proportion of the money would end up accruing as information rents. Based on this, we make the ad hoc assumption that per dollar given to the Carbon Pricing Reward Fund 0.8 dollar end up accruing as information rents, as illustrated in the diagram:

³¹Cramton and Stoft's proposal is to use an additive reward payment function, instead of the multiplicative that we consider here. The reasons why we consider the multiplicative function here are that it leads to more of the information rents accruing to low-income countries and stronger incentives for them to reduce fossil fuel subsidies and to tax carbon. Due to higher marginal utility of money in these countries, shifting information rents to them is beneficial. Moreover, in these countries, the removal of fossil fuel subsidies and the introduction of carbon taxes are very progressive in their distributional effects (Stern et al 2012).



As long as the budget of the Carbon Pricing Reward Fund is small, the information rents accruing to countries are proportional to the values of $f * \max(0, \hat{Q} - Q) * \max(0, \hat{R} - \hat{R})P$ that we obtain by substituting in the status quo values of Q and R . This is a consequence of the envelope theorem. To compute the status quo consumption-based emissions we use the available estimates and for the countries for which such estimates were not available, we use the production-based figures.

The threshold values were taken to be: $\hat{Q} = 1$ metric tons of CO₂ emissions per person per year, $\hat{R} = -\$100$ per person per year.

Concerning the 0.2 dollars that do not accrue as information rents, we make the ad hoc assumption that the multiplicative mechanism is 1.5 times as efficient at reducing emissions as the CDM. To compute the amount of oil rent redistribution, we proceed as follows: First we compute what this effect would be if all the emissions reductions achieved were through decreased oil consumption, using that in this case, the ratio of the oil rent redistribution to the climate benefit would have to be the same as we obtained in section B.3. However, we need to adjust this estimate, since part of the emissions reductions that the Carbon Pricing Reward Fund would induce could be in the form of reductions in coal and gas consumption. In fact, in 2018 the proportion of fossil fuel subsidies given to oil was 40% globally (IAEA 2019). Based on this, we make the ad hoc assumption that 40% of the emissions reductions induced by the Carbon Pricing Reward Fund are through reductions in oil consumption.³²

³²A proper estimate would require country-specific elasticity estimates and a theoretical model for how fossil fuel subsidies are set by governments. Lacking both of these elements, I have resorted to this ad hoc estimate of 40%.

B.4.5 Effects per dollar given to CEPI (the Coalition For Epidemic Preparedness Innovation)

CEPI has so far mostly been focussing on advancing vaccines against known threats. Recently, it has started to also fund new platform technologies with the potential to accelerate the development and manufacture of vaccines against previously unknown pathogens. Particularly this latter category brings benefits for all countries since future novel pathogens might cause global pandemics. Based on this, we make the assumption that the benefits of money going to CEPI accrue to all countries in proportion to their GDP. Moreover, we make the ad hoc assumption that per dollar eaching CEPI 1.5 dollars of global benefits are created.

B.4.6 Effects per dollar given to the Global Fund to Fight AIDS, TB and Malaria

The Global Fund gives grants for programs against AIDS, TB and Malaria. Allocations are calculated using a formula based on a country's disease burden and economic capacity. Of the total amount allocated to each country, at least 15 percent are made available as matching offers to encourage countries to make additional investments in their response to the three diseases. To access the incentive portion of their allocation, countries need to commit to additional co-financing investments over and above previous levels of spending.

Some countries do not take up these 1:1 matching offers, thereby revealing that they value a dollar earmarked for these health programs strictly less than a dollar with no strings attached. If there were no cross-country externalities, then we would have to conclude the aggregate global benefit³³ of a dollar given to the Global Fund is less than 1. However, taking into account the cross-country externalities weighs in favor of assuming a larger benefit-cost ratio. We make the ad hoc assumption that the actual benefit-cost ratio for a dollar given to the Global Fund is 1. Moreover, we will assume that the benefits accrue to countries in proportion to the money that they received from the Global Fund during the period 2016-2019.

B.4.7 Effects per dollar raised under CORSIA+

As explained in 7, the GPGI assumed to be selected under CORSIA+ forwards its money in equal proportions to the Fund buying up CDM Certificates and the Forest Carbon Partnership's Carbon Fund. The payoffs are therefore the average between the payoffs from the Fund buying up CDM certificates and from the Forest Carbon Partnership's Carbon Fund.

³³By "global benefit" we of course always mean the aggregate payoffs, summed over all countries. This revealed preference notion of benefits is the relevant concept when it comes to studying the incentives countries have to participate in the mechanism.

C The iterated best response algorithms

Lemma 12. *If $c_i \geq \max(\frac{f_{ki}}{f_{ki}+f_{ik}} + b_i, \theta_i^s + b_i)$ then player i 's change in payoffs from joining CORSIA+ is an increasing function of the set of other players that are participating.*

proof:

By definition, complying with the taxation obligation can be decomposed for player i into two parts: complying with the taxation obligation for the flights connecting it with j and complying with the taxation obligation for the flights connecting it with all the other countries. The costs and benefits for i associated with complying with the latter part do not depend on whether j participates or not.³⁴ Thus, to compute how j 's participation decision affects how player i 's payoffs change upon joining, we focus just on the former part, i.e. on the flights connecting i with j .

Suppose j participates. Then when player i joins, it retains $rx f_{ij}$ from the taxes collected on the flights connecting it with j .

Now suppose j does not participate. Then when country i joins, complying with the taxation obligation for the flights connecting it with j causes three effects on country i 's payoffs. Firstly, player i incurs an additional tax burden of $c_i x(f_{ij} + f_{ji})$, since these flights would otherwise not be taxed. Secondly, player i retains the amount $rx(f_{ij} + f_{ji})$. Thirdly, player i experience a change in payoffs of $(1-r)\theta_i^s x(f_{ij} + f_{ji})$ due to the increased funding for the GPII s. Fourthly, player i experience a change in payoffs of $b_i x(f_{ij} + f_{ji})$ due to the decrease in oil consumption from aviation.

Hence player i 's change in payoffs from joining always increases as a result of j participating iff the following holds:

$$rx f_{ik} \geq (-c_i + r + (1-r)\theta_i^s + b_i)x(f_{ik} + f_{ki}) \forall r$$

$$c_i \geq r(\frac{f_{ki}}{f_{ik}+f_{ki}} - \theta_i^s) + \theta_i^s + b_i \forall r$$

Suppose $\frac{f_{ki}}{f_{ik}+f_{ki}} - \theta_i^s \geq 0$. Then we obtain as a sufficient condition:

$$c_i \geq \frac{f_{ki}}{f_{ik}+f_{ki}} + b_i$$

Now suppose $\frac{f_{ki}}{f_{ik}+f_{ki}} - \theta_i^s < 0$. Then we obtain as a sufficient condition:

$$c_i \geq \theta_i^s + b_i \quad \square$$

Lemma 13. *Consider any game with a finite number m of players and with each strategy space being binary: {'not participate', 'participate'}. (Like the game induced by CORSIA+.) Suppose each player's change in payoffs from joining (i.e. from switching from 'not participate' to 'participate') is an increasing function of the set of other players that are participating. Then algorithm 1 converges almost surely.*

proof

Consider the situation at a given step n of the algorithm. We now pick a possible order in which to select players for giving them the opportunity to change their strategy. We will show that if the players are selected in this order,

³⁴This does not hold in the case of the MGF mechanism.

then after less than $2m$ steps a Nash equilibrium will be reached. Thereby we show that at any step n , the probability of reaching a Nash equilibrium in actual (random) algorithm 1 within $2m$ steps is at least m^{-2m} , which implies that algorithm 1 converges almost surely.

First, select only players who have a profitable deviation that consists of joining. Once there are no more players that have such a profitable deviation, select players who have a profitable deviation that consists of quitting. As a result, there can never be a player that will have a profitable deviation that consists of joining, since by assumption the payoff from joining can only decrease when other players quit. Thus we reach a Nash equilibrium, once there are no more players left who have a profitable deviation that consists of quitting. $\square$

Corollary 14. Suppose $c_i \geq \max(\frac{f_{ki}}{f_{ki}+f_{ik}} + b_i, \theta_i^s + b_i)$ for all players i . Then algorithm 1 converges almost surely for CORSIA+.

Corollary 15. Suppose $c_i \geq \max(\frac{f_{ki}}{f_{ki}+f_{ik}} + b_i, \theta_i^s + b_i)$ for all players i . Then algorithm 2 converges to a Nash equilibrium that is independent of the random realisation of the algorithm.

proof

First, let us show that the strategy profile reached at the end of phase 1 is independent of the random realisation of the algorithm. For this, consider two runs of the algorithm and let $(i_1, i_2, \dots, i_k), (j_1, j_2, \dots, j_l)$ be the corresponding sequences of players joining for the first and the second run of the algorithm 2, respectively. Clearly, we must have $j_1 \in \{i_1, i_2, \dots, i_l\}$, since otherwise j_i would by 12 have an incentive to join at the end of the first run of the algorithm, a contradiction. By induction, we can now show that $j_q \in \{i_1, i_2, \dots, i_l\}$ for $q = 1, \dots, k$. Hence $\{i_1, i_2, \dots, i_l\} \subset \{j_1, j_2, \dots, j_l\}$. Symmetrically, one can prove that $\{j_1, j_2, \dots, j_l\} \subset \{i_1, i_2, \dots, i_l\}$. Moreover, by an analogous argument one can now show that the set of participants at the end of phase 2 must be the same for the different realisations of the algorithm 2. $\square$

D Transposition of the MGF mechanisms to other tax bases

We have focused so far on mechanisms based on the taxation of greenhouse gas emissions from international aviation. However, the mechanism can be transposed to other tax bases. Here are two such proposals:

D.0.1 Emissions from Maritime Shipping

It was long considered to be infeasible to design a mechanism for taxing maritime shipping that could be initiated by a subset of countries, given the ease with which ships can avoid fuel taxes by fueling in jurisdictions that do not participate, and given the host of international legal constraints that apply. However, thanks to a recent concerted effort of legal scholars and economists (Heine et al (2017)) a proposal has been elaborated that navigates all these constraints and

appears to be waterproof to evasion problems. Their proposal entails that the tax liability would be situated with the consignee of the cargo for imports and the consignor of the cargo for exports.

Building on the proposal from Heine et al (2017), the MGF mechanisms could be defined completely analogously to the way we defined it for aviation emissions: The taxation obligation could entail taxing the shipping carbon emissions associated with all exports and also the shipping carbon emissions from imports from non-participating countries.

The welfare effects of taxes on the carbon emissions of shipping has been estimated by Shapiro (2016). Using these estimates, one could run simulations for the maritime-shipping-based MGF mechanisms analogous to those for the aviation-based MGF mechanism that I have reported on in the present article.

D.0.2 precious stones

The trade in gold, diamonds and other precious stones is widely recognised to contribute to violent conflict. Precious stones continue to fund armed groups, as cases in Angola and DRC show. Current international attempts to mitigate this problem have focussed on certification that aims to prevent precious stones financing violent rebel movements to enter the global market. One such mechanism is the Kimberley Process Certification Scheme. However, smuggling seems to be a problem that is difficult to overcome through the certification schemes alone.

As a complement to certification schemes, Bafilemba and Lezhnev (2015) propose reductions in taxes on gold in the Democratic Republic of the Congo. They argue: “As mines get inspected and validated, there must be a legal market ready to buy the gold that is certified as responsibly sourced. Otherwise, miners will be forced to sell through the illegal channels that benefit armed actors.”

Whilst taxing precious stones like gold in the countries where they are extracted risks increasing the funding for armed actors who extract rents in the illegal extraction and trade, taxing jewelry containing precious stones at the level of final consumption has the opposite effect: Both the illegally extracted and the legally extracted precious stones end up in jewelry that is sold legally and thus subject to consumption taxes. As a result, the prices for precious stones will fall and with it the revenues that armed groups can earn through the precious stones. This suggests that it would be beneficial to create a global mechanism that would impose taxes at the level of consumption, whilst at the same time incentivising extracting countries to lower their taxes³⁵. The precious-stones-based Simple MGF Mechanism purports to achieve exactly that. ³⁶In

³⁵Whilst tax rates on artisanally mined precious stones such as alluvial diamonds appear small, being less than 10%, Oomes and Vocke (2003) document that there are other policies that increase prices in the formal sector and thus make the informal sector more attractive. One such policy is licence fees charged on diamond traders, which also create barriers to entry and rents derived from market power. The mechanism defined below creates incentives to decrease both the taxes and these other policies.

³⁶Ng (1987) provides a further motivation for focussing on precious stones as a tax base. He shows that taxing precious stones such as diamond jewelry might be burden-free since

the definition that follows the set of all countries is denoted by I and the set of all participants to the mechanism by *Participants*.

Definition 16. The precious-stones-based Simple MGF mechanism for the fixed tax rate x , the fixed set S of GPGIs and the fixed retention rate r defines a club through the following three obligations that participating countries have to comply with:

1.1) **(the taxation obligation)**

Participating countries have to apply an ad valorem tax of rate x on all sales to consumers of goods containing precious stones and on all exports to non-participating countries of goods containing precious stones³⁷. Let M_i denote the tax revenue that the participant i collects. Before allocating or retaining money (see the allocation obligation below), each participant has to attribute part of the tax revenue to other countries according to the following procedure:

1.2) **(the attribution obligation)**

1.2.1) (trade based reattribution) Let i, j be participants and let f_{ij} be the value of exports from country i to country j of goods containing precious stones. Country j now has to attribute the amount xf_{ij} to country i .

1.2.2) (consumer sales based reattribution) Let i, j be participants and let g_{ij} be the value of sales made in country i to consumers who are residents in country j of goods containing precious stones. Country i now has to attribute the amount $x(g_{ij} - \sum_{k \in \text{Participants}} f_{ki} \frac{g_{ij}}{\sum_{l \in I} g_{il}})$ to country i .

As a result, the amount A_i of money attributed to country i is:

$$A_i = M_i - x \sum_{k \in \text{Participants}} f_{ki} + x \sum_{k \in \text{Participants}} f_{ik} - x \sum_{j \in \text{Participants}} (g_{ij} - \sum_{k \in \text{Participants}} f_{ki} \frac{g_{ij}}{\sum_{l \in I} g_{il} + f_{il}}) + x \sum_{j \in \text{Participants}} (g_{ji} - \sum_{k \in \text{Participants}} f_{kj} \frac{g_{ji}}{\sum_{l \in I} g_{jl} + f_{jl}})$$

2) **(the allocation obligation)** (analogous to the allocation obligation in the aviation-based Simple MGF mechanism³⁸, but with each player disposing of the attributed money, A_i , instead of the tax revenue collected, M_i .) Each country i can dispose of the money A_i attributed to him, according to the following rules: The country can decide on the proportion p_i of A_i that it wants to allocate to GPGIs. It can allocate the amount $p_i A_i$ in any way to the GPGIs from the set S of eligible GPGIs. Of the amount $(1 - p_i)A_i$ that country i does not allocate to GPGIs, it can retain the proportion r (which is the retention rate parameter of the mechanism). The remaining amount, $(1 - r)(1 - p_i)A_i$,

people buy them typically to give them as a gift of value. Ng's model predicts that a tax would not affect the total amount spent on precious stones. Taxes on precious stone could reap burden-free revenues of up the amount of money spent on them at the status quo, which is about \$270 billion annually.

In Ng's model the tax on precious-stones-based jewelry will not affect work incentives. In particular, the Atkinson Stiglitz theorem does not apply.

³⁷Alternatively, the ad valorem tax rate on exports to non-participants could be set at a higher rate, for example calibrated so that the average tax per carat would be the same as for the precious stones sold in goods to consumers in participating countries. Such a modification could increase participation incentives.

³⁸I have stated the definition for the transposition of the Simple MGF Mechanism. Of course, one can also transpose the MGF Mechanism with PMFs by replacing the allocation obligation accordingly.

is added to the GPGIs proportionally to the aggregate contributions made to them via the mechanism.

Here is the motivation for the definition of the attribution obligation: The first part of the attribution obligation ensures that the tax revenue is attributed to countries on the basis of the value that they add. Without any such reattribution, the countries that extract precious stones but do not have any of the other parts of the value chain on their territory, would not have an incentive to join the mechanism, even if many other countries participate.

The second part of the attribution obligation, the consumer sales' based attribution entails attributing the tax revenue xg_{ij} from sales to consumers residing in country j to that country j . The rationale for this is that it makes the mechanism conform better with the spirit of the destination principle³⁹, which asserts that "VAT or GST should be levied in the place where goods and services are consumed, not the place from which they originate". As of today, in line with the destination principle, consumers buying goods abroad can get refunds for their purchases upon leaving the country. They then have to pay taxes on these purchases in their home country. If a different version of the MGF mechanism, one without the consumer sales based attribution obligation, was adopted, then countries like China, whose residents mostly buy their luxury goods abroad, might be quite strongly adversely affected. They would have good reasons to oppose the mechanism. However, in the precious-stones-based MGF mechanism as defined above, they would have incentives to join, since by doing so they would get money attributed to them that otherwise would be attributed to the other participants.

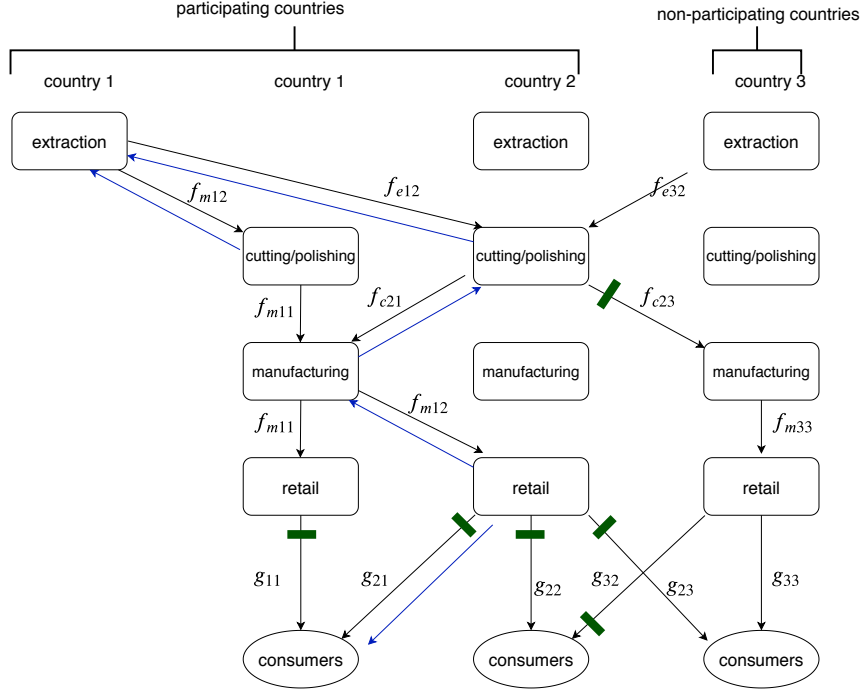
The above definition of the obligation to do consumer sales' based reattribution is designed so as to mimic what would happen if the consumption taxes were collected directly by the country in which the consumer is a resident and that country then took care of the trade-based reallocation. The mechanism only uses the aggregate trade data for each pair of countries. From that one cannot directly infer the value of the goods imported by country i that are then used to produce the goods sold in country i to consumers in country j . To estimate this value, the mechanism assumes that for each country k the exports f_{ki} to country i end up being used in goods next going to country j with a probability that is proportional to the value of the goods next going to country j , which is $g_{ik} + f_{ik}$.

The data for f_{ik} is presumably already collected by the customs authorities. To compute the reattribution amounts in accordance with the taxation obligation, one also needs for any two countries i and k that participate, the value of g_{ik} . For this, data collection could be organised as follows: consumers who are residents in country k who have bought goods containing precious diamonds in country i could register their purchase at the customs authorities before returning to their country of residence, k , just as they do today in order to get a VAT reimbursement. However, instead of getting the VAT reimbursement, they would get a certificate stating that they have bought the good in

³⁹see <https://www.ifs.org.uk/uploads/mirrleesreview/design/ch7.pdf>

a country that participates in the precious-stones-based Simple MGF Mechanism. With this certificate, the consumer would be exempted from the tax that country k would otherwise collect in line with its taxation obligation for the precious-stones-based Simple MGF Mechanism.

The diagram below illustrates the definition of the taxation obligation:



The green bars represent the taxes that participating countries would impose in accordance with the taxation obligation. The blue arrows represent the reattributions.

Sketch of a model period 1: countries choose whether how to participate in the precious-stones-based Simple MGF Mechanism (i.e. if they participate they choose how much of the tax revenue attributed to them they allocate and how they allocate this money to the GPIs recognised as eligible in the mechanism.

period 2: each country decides on how much precious stones to extract.

period 3: A Walrasian equilibrium emerges (treating the quantities extracted in period 2 as endowments).

In such a model the following effect could be quantified: When a country (for example India) that imports rough diamonds joins the mechanism, it benefits in two ways: Firstly, it gets money attributed from participating countries that it exports to. Secondly since it causes more money to be attributed to participating rough diamond producing countries, it incentivises the latter to decrease taxes to increase diamond production, thus decreasing the world market price of rough diamonds, which might in turn benefit India. Thus the participation

incentives might be larger than one might first expect.

With a fully specified version of this model the simulations that I have run for the aviation-based MGF mechanisms could be rerun.

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